

CHANGECARS

Backend & API Integration Guide

How the platform works, and every API endpoint explained in simple English for app developers

Written for

- Mobile app developers (Android, iOS, Flutter, React Native)
- Web frontend developers (customer site, dealer portal, admin panel)
- Backend developers, QA testers and technical project managers

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1. About this guide

ChangeCars is an online marketplace for buying and selling vehicles in South Africa. Customers search new and used cars, contact dealers, ask for quotes, compare vehicles, use finance calculators and sell or value their own car. Dealers manage their stock, leads and staff, and bid on cars that customers want to sell. The ChangeCars team manages dealers, listings, content and enquiries.

This guide explains the backend that powers all of this. It has two goals:

- Explain **how the backend works**: the main ideas, rules and flows, in simple English.
- Give **every API endpoint** with its access rules, parameters and request fields, so an app developer can connect a mobile app, website, dealer portal or admin panel without reading the backend code.

1.1 Who should read what

Reader	Read these chapters
Mobile app developer (customer app)	3, 4, 5, then 6.1 to 6.8, chapter 7 and the Public and customer part of chapter 8
Web developer (public website)	Same as the mobile developer, plus 6.11 (content and SEO)
Dealer portal / dealer app developer	3, 4, 5, 6.9 and the Dealer portal part of chapter 8
Admin panel developer	3, 4, 5, 6.10 and the Admin part of chapter 8
Backend developer, DevOps	Everything, especially chapter 2 and chapter 9
QA tester	Chapters 4 to 7 and appendix A (error codes)

1.2 Words used in this guide

Word	Meaning
Visitor	Someone using the app or website without signing in.
Customer	A signed-in user who buys or sells a car.
Dealer / dealership	A car dealer business registered on ChangeCars. It can have several branches.
Dealer member (staff)	A user who works for a dealership. Has a role (owner, manager, sales, staff) and permissions.
Admin, super admin	People from the ChangeCars team. A super admin can also manage other admins.
Vehicle / listing	A car offered for sale by a dealer.
Catalogue	The list of makes, models, generations, variants and their specifications, for example BMW > 3 Series > G20 > 320i.
Enquiry	A message or request from a customer: contact dealer, quote, trade-in, finance, and so on.
Lead	A potential sale in a dealer's CRM. Most enquiries create a lead automatically.
Sell request	A customer's request to value or sell their own car.
Valuation	An estimated trade value for the customer's car, automatic or done by the team.
Bidding session	A time window in which dealers can make offers on a customer's car.
Offer, counter-offer	A dealer's price for the customer's car; the customer can answer with a higher counter-offer.
Deal	Created when a customer accepts an offer.
Slug	A readable id used in URLs, for example 2022-bmw-3-series-320i-x7k2pq.
Access token, refresh token	The two tokens the app receives after sign-in. See chapter 5.
Worker	A background process that sends emails, notifications and alerts and processes images.

2. The system at a glance

2.1 Technology

Part	Technology	Why
API server	Node.js 20+, NestJS 11, TypeScript	Clear module structure, validation, OpenAPI documentation
Database	PostgreSQL with Prisma 7	The single source of truth for all business data
Cache and limits	Redis	Fast cache, shared rate limits, view counters across many servers
File storage	S3-compatible storage (AWS S3, Cloudflare R2, MinIO)	Vehicle photos and documents never live on the server disk
Background work	Worker process with a database outbox	Emails, notifications, alerts and image processing never slow down a request
Images	sharp	Photos are resized to 320, 800 and 1600 px WebP; location data is removed

2.2 How a request travels

```

App / website
|
v
CDN + firewall (caches public, anonymous GET responses)
|
v
Load balancer / Nginx
|
v
API servers (1..N, all identical, no local state)
|-----> PostgreSQL (primary, plus optional read replica for search)
|-----> Redis (cache, rate limits, view counters)
|-----> Object storage (photos, documents)

Worker servers (1..N) -> read the outbox table -> send email/SMS/in-app notifications,
                        process photos, run valuations, expire offers, release reservations

```

Every API server is the same and keeps nothing in memory between requests. So any server can answer any request, and more servers can be added when traffic grows. Public pages such as search results and vehicle pages are cacheable by the CDN, so most browsing traffic never reaches the API.

2.3 The two processes

- **API** (`node dist/main.js`): answers HTTP requests.
- **Worker** (`node dist/worker.js`): does background work. When something happens (for example a new enquiry), the API saves an event in the database in the same step as the change. The worker picks it up and sends the notifications. If sending fails, the worker retries with growing delays. This is why some results, such as a valuation or an email, arrive a few seconds after the request.

2.4 Modules

Module	What it does
Auth	Registration, login, tokens, passwords
Customers	Profile, dashboard, favourites, saved searches, recently viewed, privacy
Dealers	Dealer registration and approval, profile, branches, staff and permissions
Catalogue	Makes, models, generations, variants, specifications, categories, features
Vehicles	Dealer inventory, listing lifecycle, photos, public vehicle pages
Search	Filters, sorting, collections, facets, saved-search alerts
Enquiries	All customer request forms and conversations
CRM	Dealer leads, stages, assignment, contact history
Selling	Sell requests and valuations
Bidding	Dealer bidding, offers, counter-offers, acceptance, deals
Finance	Repayment and affordability calculators
Content	Articles, videos, podcasts, FAQs, static pages, promotions
Newsletter	Subscriptions
Notifications	In-app notifications, email, SMS, user preferences
Dealer dashboard	Summary numbers for dealers
Admin	Users, statistics, audit log, background job health
SEO	Sitemaps and structured data
Health	Liveness and readiness checks

3. Getting started

3.1 Base URL and documentation

Environment	Base URL	Notes
Local (developer machine)	<code>http://localhost:4000/api/v1</code>	Run the backend yourself (see the backend README)
Staging	<code>https://<staging-host>/api/v1</code>	Address provided by the backend team
Production	<code>https://<production-host>/api/v1</code>	Address provided by the backend team

- **Interactive docs (Swagger UI):** `/docs`. You can try every endpoint in the browser. Click Authorize and paste an access token to call protected endpoints.
- **OpenAPI file:** `/docs/openapi.json`. Use it to generate a typed API client, for example with openapi-generator (Kotlin/Retrofit, Swift, Dart) or openapi-typescript (TypeScript).
- **Health checks:** `/health/live` and `/health/ready` (outside `/api/v1`).

Swagger in production

Interactive docs can be switched off in production with `SWAGGER_ENABLED=false`. Ask the backend team for a copy of `openapi.json` if it is not exposed.

3.2 Test accounts (demo data)

When the backend team loads the demo data, these accounts exist. All use the password **Password123** except the admin, whose password is chosen when the data is loaded.

Role	Email	Use it to test
Super admin	<code>admin@changecars.co.za</code>	Admin panel
Dealer owner	<code>owner@demo-dealer.co.za</code>	Dealer portal with all permissions
Dealer sales staff	<code>sales@demo-dealer.co.za</code>	Dealer portal with limited permissions
Customer	<code>customer@example.com</code>	Customer app

3.3 Your first calls

1. Search vehicles (public, no login):

```
curl "http://localhost:4000/api/v1/vehicles?make=toyota&fuelType=DIESEL&sort=price-asc"
```

2. Create an account. The response contains the tokens:

```
curl -X POST http://localhost:4000/api/v1/auth/register \
-H "Content-Type: application/json" \
-d '{"email":"sipho@example.com","password":"Secret123","firstName":"Sipho",
    "lastName":"Dlamini","phone":"+27 82 555 0101","acceptTerms":true}'
```

3. Call a protected endpoint with the access token:

```
curl http://localhost:4000/api/v1/me/dashboard \
-H "Authorization: Bearer <accessToken>"
```

4. Rules every request follows

4.1 Data format

Topic	Rule
Format	JSON in and out, UTF-8. Send Content-Type: application/json with a body. Maximum body size 1 MB.
Field names	camelCase, for example firstName , vehicleId .
Ids	UUID strings, for example 01a1124c-58d2-7a3e-9b1f-2c7d5e0f4a11 . Vehicles, dealers, articles, promotions and media also have a slug for nice URLs.
Dates and times	ISO 8601 in UTC, for example 2026-10-06T17:26:35.115Z . Convert to local time in the app (South Africa is UTC+2). Send dates in the same format.
Money	Whole South African rand as integers (no cents), for example price 459900 means R459 900. Only finance calculator results contain cents.
Distances and sizes	Mileage in km, engine in cc, power in kW, torque in Nm, dimensions in mm, fuel consumption in litres per 100 km.
Fixed values (enums)	UPPER_CASE strings, for example PUBLISHED , GAUTENG . All values are listed in chapter 7.
Unknown fields	Rejected. Sending a field the API does not know returns 400 VALIDATION_FAILED. This protects against typos.
Text cleaning	Leading and trailing spaces are removed. Emails are stored in lower case.

4.2 Lists and paging

Every list endpoint is paged. Send **page** (starts at 1) and **pageSize** (default 20, maximum 100; vehicle search maximum 50). The response always looks like this:

```
{
  "data": [
    "... items ..."
  ],
  "meta": {
    "page": 1,
    "pageSize": 20,
    "total": 134,
    "pageCount": 7
  }
}
```

Use **meta.pageCount** to know when to stop loading more (infinite scroll), and **meta.total** to show result counts.

4.3 Errors

Every error has the same shape. Your app should decide what to do based on **code** (stable, machine-readable) and may show **message** to the user. Always log **requestId**; the backend team can find the exact request with it.

Example: wrong password (401)

```
{
  "statusCode": 401,
  "code": "INVALID_CREDENTIALS",
  "message": "Email or password is incorrect",
  "requestId": "dfd5c749-d486-4e7b-a9f5-0386b66f882d",
  "timestamp": "2026-10-06T17:46:21.048Z",
  "path": "/api/v1/auth/login"
}
```

Example: validation error (400). details lists every problem

```
{
  "statusCode": 400,
  "code": "VALIDATION_FAILED",
  "message": "Request validation failed",
  "details": [
    "email must be an email",
    "Password must be 8-128 characters and contain at least one letter and one number"
  ],
  "requestId": "4face2e1-0957-4f95-96e7-8a4c1f819dac",
  "timestamp": "2026-10-06T17:46:21.056Z",
  "path": "/api/v1/auth/register"
}
```

HTTP status	Meaning	What the app should do
400	The request is wrong (validation, bad values)	Show the field errors from details. Do not retry.
401	Not signed in, or the access token expired	Refresh the token once, then retry. If refresh fails, go to the sign-in screen.
403	Signed in but not allowed	Hide the action, or explain why (for example the dealership is not approved yet).
404	Not found (or you may not see it)	Show a not-found screen.
409	Conflict with the current state	Reload the data and show the message, for example 'This offer has expired'.
422	Valid request but a business rule is not met	Show the message, for example 'Upload at least one photo first'.
429	Too many requests	Wait a few seconds and try again with a growing delay.
500 / 503	Server problem	Show a friendly error and allow retry. Report the requestId.

All error codes are listed in appendix A.

4.4 Request id

Every response has an **X-Request-Id** header. You may also send your own **X-Request-Id** (8 to 100 letters, digits, dot, dash or underscore) to follow one user action across app logs and server logs.

4.5 Rate limits

Limits protect the platform against abuse. They count requests per IP address per minute:

Endpoints	Limit
Most endpoints	300 requests per minute (configurable)
Login, register, reset password	10 per minute
Forgot password, dealer registration, newsletter subscribe	5 per minute
Refresh token	30 per minute
Enquiry forms and sell requests	10 per minute

4.6 Caching

Public, anonymous GET responses (search, vehicle pages, catalogue, content, dealers, sitemaps) include **Cache-Control: public, s-maxage=...**. The CDN and the app's HTTP client may cache them. Public data may therefore be up to about one minute old (catalogue up to one hour). Responses to signed-in requests always carry **Cache-Control: private, no-store**. Before any money or status action (enquiry, offer, accept) the server always checks the latest data, so cached data can never cause a wrong transaction.

4.7 Safe retries (Idempotency-Key)

Accepting an offer is important and must never happen twice. Send a header **Idempotency-Key: <uuid>** with a new random UUID for each user tap. If the network fails and you retry with the **same key**, the server returns the first result instead of doing the action again. Keys are kept for 24 hours. Using the same key with a different request body returns 422 IDEMPOTENCY_KEY_REUSED.

4.8 Editing safely (versions)

Vehicles and offers have a **version** number. When you edit, you can send **expectedVersion** with the version you loaded. If someone else changed the record in the meantime, the server returns 409 STALE_VERSION and you should reload and show the latest data.

5. Sign-in, tokens and permissions

5.1 The two tokens

Token	Lifetime	How to use it
Access token (JWT)	15 minutes	Send on every protected request: Authorization: Bearer <accessToken>
Refresh token	30 days, single use	Send to POST /auth/refresh to get a new pair. Each refresh token works only once.

Register and login both return the same response:

POST /auth/register response (201)

```
{
  "tokenType": "Bearer",
  "accessToken": "<access token>",
  "expiresIn": 900,
  "refreshToken": "<refresh token>",
  "refreshTokenExpiresAt": "2026-11-05T17:46:20.763Z",
  "user": {
    "id": "01a11252-d4d1-743d-bbbb-eca9ba449aaf",
    "email": "docs.1791308780443@example.com",
    "firstName": "Sipho",
    "lastName": "Dlamini",
    "phone": "+27 82 555 0101",
    "role": "CUSTOMER"
  }
}
```

5.2 Keeping the user signed in

1. Save both tokens in secure storage: iOS Keychain / Android Keystore (for example flutter_secure_storage or expo-secure-store). On the web, keep the access token in memory and protect the refresh token carefully.
2. Add the access token to every request.
3. When a request returns **401** with code **INVALID_TOKEN**, call **POST /auth/refresh** with the refresh token.
4. Save the **new** access and refresh tokens from the response (the old refresh token is now dead), then retry the original request once.
5. If refresh returns 401 or 403, clear the tokens and show the sign-in screen.

Important: only one refresh at a time

If two requests fail together, refresh once and let the other requests wait for the result. Using the same refresh token twice is treated as token theft: the server signs the user out on all devices (code **INVALID_REFRESH_TOKEN**). You can also refresh a little before expiresIn runs out.

```
// Pseudo-code for an HTTP interceptor
let refreshing = null;
async function request(options) {
  let res = await send(options, accessToken);
  if (res.status === 401 && res.body.code === 'INVALID_TOKEN') {
    refreshing ??= refreshTokens().finally(() => (refreshing = null)); // one refresh only
    const ok = await refreshing;
    if (!ok) return goToSignIn();
    res = await send(options, accessToken);
  }
  return res;
}
```

5.3 Who is signed in

After sign-in, call **GET /auth/me**. It returns the user and, for dealer staff, the dealership and member role. Use **role** to decide which app or portal to show. Dealer apps can call **GET /dealer/me** to get the exact permission list:

```
GET /dealer/me (dealer owner)
{
  "userId": "01a11251-6736-77aa-acbe-25e5a7fa3abf",
  "memberId": "01a11251-6744-739d-bc77-d62986615b39",
  "dealerId": "01a11251-673d-7396-8b37-34ce4821d788",
  "dealerStatus": "APPROVED",
  "role": "OWNER",
  "branchId": null,
  "permissions": [
    "DEALER_PROFILE_MANAGE",
    "BRANCHES_MANAGE"
  ],
  "wideScope": true
}
```

5.4 Passwords

- Passwords need 8 to 128 characters with at least one letter and one number.
- **Forgot password:** **POST /auth/forgot-password** always answers 202, even for unknown emails (so nobody can test which emails exist). The user gets an email with a link to `<website>/reset-password?token=...`. The app or website reads the token and calls **POST /auth/reset-password**. The link works for 60 minutes. After a reset, all devices are signed out.
- **Change password** (signed in): **POST /auth/change-password**. Other devices are signed out; this one stays signed in.
- **Staff invitations:** new dealer staff and new admins receive the same reset-password link (with `&invite=1`), valid for 7 days, to choose their first password.
- **Sign out:** **POST /auth/logout** (this device) or **POST /auth/logout-all** (all devices). Then delete the stored tokens.

5.5 Platform roles

Role	Can do
Visitor (not signed in)	Search and view vehicles, read content, use calculators, send enquiries, start a valuation, register
CUSTOMER	Everything a visitor can, plus dashboard, favourites, saved searches, alerts, enquiry tracking, sell requests and offers
DEALER	Dealer portal according to the member's dealership permissions
ADMIN	Admin panel: dealers, listings, catalogue, content, enquiries, bidding, users, statistics
SUPER_ADMIN	Everything an admin can, plus creating admins and changing user roles

5.6 Dealer roles and permissions

Each dealership member has a role with default permissions. An owner or manager can give extra permissions to a member, but never a permission they do not have themselves. Members of a branch (sales, staff) only see vehicles, enquiries and leads of their own branch.

Permission	Allows	Owner	Manager	Sales	Staff
DEALER_PROFILE_MANAGE	Edit dealership profile, upload verification documents	Yes	-	-	-
BRANCHES_MANAGE	Add and edit branches	Yes	Yes	-	-
STAFF_MANAGE	Create, change and disable staff accounts	Yes	Yes	-	-
INVENTORY_VIEW	See the vehicle inventory	Yes	Yes	Yes	Yes
INVENTORY_MANAGE	Create and edit vehicles, photos, submit, reserve, sell, suspend, archive	Yes	Yes	-	-
INVENTORY_PUBLISH	Publish approved vehicles	Yes	Yes	-	-
ENQUIRIES_VIEW	See customer enquiries	Yes	Yes	Yes	-
ENQUIRIES_RESPOND	Reply to customer enquiries	Yes	Yes	Yes	-
LEADS_VIEW_ALL	See all leads (otherwise only leads assigned to me)	Yes	Yes	-	-
LEADS_MANAGE_ALL	Change any lead	Yes	Yes	-	-
LEADS_MANAGE_ASSIGNED	Change leads assigned to me	Yes	Yes	Yes	Yes
LEADS_ASSIGN	Assign leads to staff	Yes	Yes	-	-
BIDDING_VIEW	See vehicles open for bidding and my offers	Yes	Yes	Yes	-
BIDDING_PARTICIPATE	Submit, revise and withdraw offers	Yes	Yes	-	-
REPORTS_VIEW	See reports	Yes	Yes	-	-

Dealership status matters

A PENDING dealership can sign in, edit its profile, add branches and staff and create draft vehicles, but cannot submit listings for review, publish or bid until an admin approves it (code DEALER_NOT_APPROVED). A SUSPENDED or REJECTED dealership can only view its profile and dashboard (code DEALER_INACTIVE).

6. Feature guides

This chapter explains each feature as a list of steps with the endpoints to call. Full details of every endpoint are in chapter 8.

6.1 Search and vehicle pages

GET /vehicles returns published and reserved vehicles from approved dealers. All filters are optional and can be combined. Filters with several values take a comma-separated list; a vehicle matches if it has any of the values.

Parameter	Meaning	Example
q	Words in the title (all words must match)	q=hilux legend
make	Make slugs	make=bmw,audi
model	Model slugs. Use make:model to limit a model to one make	make=bmw,toyota&model=toyota:hilux
variant	Variant slugs	variant=320i-2022
condition	NEW, USED, DEMO	condition=USED
category	Category slugs (body type, EV, hybrid...)	category=suvs,double-cabs
fuelType / transmission / drivetrain	Values from chapter 7	fuelType=DIESEL,PETROL
province / city	Location	province=GAUTENG
colour	Colours	colour=White,Black
dealer / branchId	One dealer (slug) or branch (id)	dealer=sandton-auto
minPrice / maxPrice	Price range in rand	maxPrice=400000
minYear / maxYear	Model year range	minYear=2020
minMileage / maxMileage	Mileage range in km	maxMileage=80000
minEngineCc / maxEngineCc, minPowerKw / maxPowerKw	Engine size and power	minPowerKw=100
seats	Seat counts, 8+ means eight or more	seats=7,8+
onSpecial / featured / availableOnly	Only specials, only featured, hide reserved	availableOnly=true
lat, lng, radiusKm	Vehicles within a distance of a point	lat=-26.1&lng=28.05&radiusKm=25
collection	Ready-made lists: hot-sellers, specials, featured, new-arrivals, budget, student, bakkies, electric	collection=bakkies
sort	recent, oldest, price-asc, price-desc, mileage-asc, mileage-desc, year-desc, year-asc, popular	sort=price-asc
page / pageSize	Paging (pageSize max 50)	page=2&pageSize=20

GET /vehicles?make=toyota&fuelType=DIESEL&sort=price-asc&pageSize=1 (one result shown)

```
{
  "data": [
    {
      "id": "01a11251-678a-706f-a432-6e6ba34717d4",
      "slug": "2021-toyota-hilux-2-4-gd-6-raider-xtra-cab-demo3",
      "title": "2021 Toyota Hilux 2.4 GD-6 Raider Xtra Cab",
      "condition": "USED",
      "status": "PUBLISHED",
      "year": 2021,
      "mileage": 92000,
      "price": 409900,
      "specialPrice": null,
```

```

"isSpecial": false,
"isFeatured": false,
"transmission": "MANUAL",
"fuelType": "DIESEL",
"drivetrain": "FOUR_X_TWO",
"colour": "Silver",
"engineCapacityCc": 2393,
"powerKw": 110,
"cylinders": 4,
"seats": 4,
"doors": 2,
"province": "GAUTENG",
"city": "Sandton",
"primaryImageUrl": null,
"imageCount": 0,
"viewCount": 42,
"publishedAt": "2026-10-03T17:44:47.240Z",
"reservedUntil": null,
"make": {
  "id": "01a11251-658f-716c-9533-7f4bc1ad7b63",
  "name": "Toyota",
  "slug": "toyota"
},
"model": {
  "id": "01a11251-6590-7519-b838-1d0c395b06c6",
  "name": "Hilux",
  "slug": "hilux"
},
"variant": {
  "id": "01a11251-659a-753b-800c-4da549424b45",
  "name": "2.4 GD-6 Raider Xtra Cab",
  "slug": "2-4-gd-6-raider-xtra-cab-2021"
},
"dealer": {
  "id": "01a11251-673d-7396-8b37-34ce4821d788",
  "name": "Sandton Auto",
  "slug": "sandton-auto",
  "logoUrl": null,
  "rating": 4.6,
  "plan": "GOLD"
},
"branch": {
  "id": "01a11251-6740-70e9-af43-3e258ad6ec2d",
  "name": "Sandton Auto - Sandton",
  "city": "Sandton",
  "province": "GAUTENG"
},
"categories": [
  {
    "slug": "king-cabs",
    "name": "King Cabs"
  }
],
"promotion": null,
"availability": "AVAILABLE",
"effectivePrice": 409900,
"discount": null
}
... (shortened)

```

- **Show the price:** use `effectivePrice`. If `discount` is not null, show the old `price` crossed out (vehicle on special).
- **Show the status:** use `availability` (AVAILABLE, RESERVED, SOLD...). See chapter 7.
- **Images:** `primaryImageUrl` is the medium-size main photo for list cards.
- **Filter counts:** `GET /vehicles/facets` accepts the same filters and returns counts per make, fuel, transmission, province, condition and category, plus min/max price, year and mileage for range sliders.
- **Specials and hot sellers:** `GET /vehicles/specials` and `GET /vehicles/hot-sellers` (sorted by popularity).

Vehicle page

GET /vehicles/{slugOrId} accepts the slug or the id. It returns photos in three sizes, full specifications, features (each with STANDARD or OPTIONAL and where it comes from), dealer and branch (with opening hours), and an example finance repayment. Signed-in users automatically get the vehicle added to their recently viewed list.

GET /vehicles/{slug} (shortened)

```
{
  "id": "01a11251-678a-706f-a432-6e6ba34717d4",
  "slug": "2021-toyota-hilux-2-4-gd-6-raider-xtra-cab-demo3",
  "title": "2021 Toyota Hilux 2.4 GD-6 Raider Xtra Cab",
  "condition": "USED",
  "status": "PUBLISHED",
  "year": 2021,
  "mileage": 92000,
  "price": 409900,
  "specialPrice": null,
  "isSpecial": false,
  "isFeatured": false,
  "transmission": "MANUAL",
  "fuelType": "DIESEL",
  "drivetrain": "FOUR_X_TWO",
  "colour": "Silver",
  "engineCapacityCc": 2393,
  "powerKw": 110,
  "cylinders": 4,
  "seats": 4,
  "doors": 2,
  "province": "GAUTENG",
  "city": "Sandton",
  "primaryImageUrl": null,
  "imageCount": 0,
  "viewCount": 42,
  "publishedAt": "2026-10-03T17:44:47.240Z",
  "reservedUntil": null,
  "make": {
    "id": "01a11251-658f-716c-9533-7f4bc1ad7b63",
    "name": "Toyota",
    "slug": "toyota"
  },
  "model": {
    "id": "01a11251-6590-7519-b838-1d0c395b06c6",
    "name": "Hilux",
    "slug": "hilux"
  },
  "variant": {
    "id": "01a11251-659a-753b-800c-4da549424b45",
    "name": "2.4 GD-6 Raider Xtra Cab",
    "slug": "2-4-gd-6-raider-xtra-cab-2021",
    "specification": {
      "variantId": "01a11251-659a-753b-800c-4da549424b45",
      "engine": "2.4L 4-cylinder turbo diesel",
      "engineCapacityCc": 2393,
      "cylinders": 4,
      "powerKw": 110,
      "torqueNm": 400,
      "transmission": "MANUAL",
      "gears": 6,
      "drivetrain": "FOUR_X_TWO",
      "fuelType": "DIESEL",
      "fuelConsumptionL100": 7.1,
      "co2GKm": null,
      "seats": 4,
      "doors": 2,
      "lengthMm": 5325,
      "widthMm": 1855,
      "heightMm": 1800,

```

```

    "wheelbaseMm": null,
    "bootLitres": null,
    "kerbWeightKg": null,
    "fuelTankLitres": null,
    "topSpeedKmh": null,
    "zeroTo100Sec": null,
    "batteryKwh": null,
    "electricRangeKm": null,
    "warranty": "3 years / 100 000 km",
    "servicePlan": "6 services / 90 000 km",
    "safetyRating": null,
    "extra": null,
    "updatedAt": "2026-10-06T17:44:46.746Z"
  },
  "dealer": {
    "id": "01a11251-673d-7396-8b37-34ce4821d788",
    "name": "Sandton Auto",
    "slug": "sandton-auto",
    "logoUrl": null,
    "rating": 4.6,
    "plan": "GOLD",
    "phone": "+27 11 000 0000",
    "email": "sales@sandtonauto.co.za",
    "website": null,
    "address": "1 Rivonia Road, Sandton",
    "city": "Sandton",
    "province": "GAUTENG"
  },
  "branch": {
    "id": "01a11251-6740-70e9-af43-3e258ad6ec2d",
    "name": "Sandton Auto - Sandton",
    "address": "1 Rivonia Road, Sandton",
    "city": "Sandton",
    "province": "GAUTENG",
    ... (shortened)
  }
}

```

- **canEnquire** is false for sold vehicles: hide the enquiry buttons.
- **finance** is an example with 10% deposit, 72 months and 11.75% interest. Let users change it with the finance calculator.
- Related calls: **/vehicles/{id}/similar**, **/vehicles/{id}/dealer-vehicles**, **/vehicles/{id}/market-price** (average price of comparable cars) and **/vehicles/compare?ids=a,b,c** (2 to 4 vehicles).
- For SEO the website can embed **/seo/vehicles/{slug}/structured-data** (schema.org JSON-LD).

6.2 Catalogue (makes, models, variants)

Use the catalogue for make/model dropdowns, the new-vehicle flow and specification pages. It changes rarely, so cache it in the app for a day.

1. **GET /catalogue/makes** for the list of makes.
2. **GET /catalogue/makes/{makeSlug}/models** for the models of a make.
3. **GET /catalogue/makes/{makeSlug}/models/{modelSlug}** for generations and variants with specifications and list prices (new vehicles).
4. **GET /catalogue/variants/{id}** for one variant with all specifications and features; **GET /catalogue/variants/compare?ids=...** to compare variants.
5. To ask for a price, send a quote request: **POST /enquiries/quote** (see 6.4).

GET /catalogue/makes (two of the results)

```

[
  {
    "id": "01a11251-6568-72ff-b69c-5aa58829742d",
    "name": "BMW",
    "slug": "bmw",
    "logoUrl": null,

```

```

    "country": "Germany"
  },
  {
    "id": "01a11251-658f-716c-9533-7f4bc1ad7b63",
    "name": "Toyota",
    "slug": "toyota",
    "logoUrl": null,
    "country": "Japan"
  }
]

```

GET /catalogue/categories returns the vehicle categories (hatchbacks, sedans, SUVs, crossovers, coupés, convertibles, bakkies, super cabs, king cabs, double cabs, MPVs, motorbikes, electric, hybrid). Use their slugs in the category search filter.

6.3 Favourites, saved searches, alerts and history

Feature	Endpoints	Notes
Save a vehicle	PUT /me/favourites/{vehicleId}, DELETE /me/favourites/{vehicleId}	Safe to call twice. Saved on the server, so it works on every device.
Heart icons	GET /me/favourites/ids	Load once after sign-in to mark saved vehicles in lists.
Favourites list	GET /me/favourites	Paged, with current availability.
Saved searches	GET/POST /me/saved-searches, PATCH/DELETE /me/saved-searches/{id}	Store the same filters as GET /vehicles in criteria. Up to 25 per user.
Alerts	Automatic	With alertsEnabled=true the user is notified about new matching vehicles, price drops and vehicles that become available again.
Recently viewed	GET /me/recently-viewed, DELETE /me/recently-viewed	Filled automatically when a signed-in user opens a vehicle page. Last 50 kept.

POST /me/saved-searches body

```

{
  "name": "Diesel double cabs under R600k",
  "criteria": {
    "fuelType": "DIESEL",
    "category": "double-cabs",
    "maxPrice": 600000,
    "province": "GAUTENG"
  },
  "alertsEnabled": true
}

```

6.4 Enquiries and customer requests

All forms share four contact fields: **name**, **email**, **phone** and **consent**. The consent must be **true** (the user agrees to be contacted, as required by POPIA). Login is optional, but when the user is signed in, send the token so the request appears in their dashboard.

Endpoint	Use for	Extra fields	Who handles it
POST /enquiries/vehicle	Contact dealer about a vehicle	vehicleId, message, interestedInFinance, hasTradeln	The vehicle's dealer (lead created)
POST /enquiries/test-drive	Book a test drive	vehicleId, preferredAt, message	The vehicle's dealer (lead created)
POST /enquiries/quote	Quote for a vehicle	makeld, modelId, variantId, dealerId, province, requirements	Chosen dealer, else the ChangeCars team
POST /enquiries/beat-my-quote	Beat an existing quote	desiredVehicle, quotedPrice, quotingDealer, quoteDetails, province	ChangeCars team

Endpoint	Use for	Extra fields	Who handles it
POST /enquiries/trade-in	Trade in while buying	vehicleId or desiredVehicle, tradeIn {make, model, year, mileage, condition...}	Vehicle's dealer, else the team
POST /enquiries/concierge	Concierge service	services[], budget, province, notes	ChangeCars team
POST /enquiries/help-me-find	Find a vehicle for me	makeId, modelId, vehicleType, budgetMin, budgetMax, minYear, maxMileage, province, preferences	ChangeCars team
POST /enquiries/finance	Finance interest	vehicleId, vehiclePrice, deposit, termMonths, monthlyIncome, employmentStatus	Vehicle's dealer, else the team
POST /enquiries/insurance	Insurance quote	vehicleId, vehicleDescription, coverType, province	ChangeCars team
POST /enquiries/contact	Contact us	subject, message	ChangeCars team

POST /enquiries/vehicle body

```
{
  "vehicleId": "01a11251-678a-706f-a432-6e6ba34717d4",
  "name": "Sipho Dlamini",
  "email": "docs.1791308780443@example.com",
  "phone": "+27 82 555 0101",
  "consent": true,
  "message": "Is this still available?",
  "interestedInFinance": true
}
```

Response (201). Show the reference to the user

```
{
  "id": "01a11252-d6c9-720c-8ab6-34aac7a1153d",
  "reference": "ENQ-CEGAWFE4",
  "type": "VEHICLE",
  "status": "NEW",
  "createdAt": "2026-10-06T17:46:21.257Z",
  "duplicate": false
}
```

- If the same email sends the same request again within 10 minutes, the first enquiry is returned with **duplicate: true**. Show it as a success.
- Enquiries for sold, suspended or archived vehicles are refused with 409 **VEHICLE_NOT_AVAILABLE**.
- The customer receives a confirmation email. Signed-in customers follow replies in **GET /me/enquiries**, can reply with **POST /me/enquiries/{id}/reply** and cancel with **POST /me/enquiries/{id}/cancel**.

6.5 Finance calculators

POST /finance/repayment body

```
{
  "vehiclePrice": 400000,
  "deposit": 40000,
  "termMonths": 72,
  "annualInterestRate": 11.75,
  "balloonPercent": 0
}
```

Response

```
{
  "vehiclePrice": 400000,
  "deposit": 40000,
  "financeAmount": 360000,
  "balloonAmount": 0,
  "termMonths": 72,
```

```

"annualInterestRate": 11.75,
"monthlyRepayment": 6991.35,
"totalRepayment": 503377.55,
"totalInterest": 143377.55
}

```

POST /finance/affordability takes monthlyIncome, monthlyExpenses, deposit, termMonths, annualInterestRate and optional balloonPercent. It returns the maximum monthly repayment (capped at 30% of income and at disposable income) and an estimated vehicle budget, with a list of assumptions to show under the result.

6.6 Sell or value my car

This is the full journey from 'what is my car worth?' to an accepted dealer offer.

1. **Submit the car.** **POST /sell-requests** with contact details, province and vehicle details. Send catalogue ids (makeId, modelId, variantId) when possible; otherwise makeName and modelName. type=VALUATION for a value only, SELL for dealer offers.
2. **Valuation.** The worker calculates an automatic valuation within seconds when there is enough market data. The user gets a valuation.ready notification. Without enough data the request goes to UNDER_REVIEW and the team adds a manual valuation.
3. **Photos (optional, signed-in users).** **POST /me/sell-requests/{id}/images/upload-url**, then PUT the file to the returned URL, then confirm with **POST /me/sell-requests/{id}/images**.
4. **Ask for offers.** **POST /me/sell-requests/{id}/request-offers**. The team opens a bidding session for approved dealers.
5. **Receive offers.** Dealers bid. The user gets offer.new notifications. Show offers from **GET /me/offers** or **GET /me/sell-requests/{id}**: dealer name and rating, amount, terms and expiry time.
6. **Answer offers.** Accept with **POST /me/offers/{id}/accept** (send an Idempotency-Key), reject with **POST /me/offers/{id}/reject**, or counter with a higher amount using **POST /me/offers/{id}/counter** (when the session allows it).
7. **Deal.** After acceptance all other offers become SUPERSEDED, the request becomes OFFER_ACCEPTED, a deal is created and the dealer contacts the customer.

POST /sell-requests body

```

{
  "type": "SELL",
  "name": "Sipho Dlamini",
  "email": "docs.1791308780443@example.com",
  "phone": "+27 82 555 0101",
  "consent": true,
  "province": "GAUTENG",
  "makeName": "Volkswagen",
  "modelName": "Polo",
  "year": 2019,
  "mileage": 85000,
  "condition": "GOOD"
}

```

Response (201)

```

{
  "id": "01a11252-d720-7343-9950-d760cb28f219",
  "reference": "SELL-UG84JFQ4",
  "type": "SELL",
  "status": "SUBMITTED",
  "createdAt": "2026-10-06T17:46:21.344Z"
}

```

Accepting an offer can fail for good reasons

The server checks the offer again at the moment of acceptance. Show the message for these codes and reload the offers:

OFFER_EXPIRED, OFFER_WITHDRAWN, OFFER_REJECTED, OFFER_SUPERSEDED, ALREADY_AWARDED, BIDDING_CANCELLED. Only one offer per car can ever be accepted, even if two taps happen at the same time.

6.7 Notifications

- Notifications are delivered in the app (in-app list), by email and, when a provider is configured, by SMS.
- Show a badge with `GET /me/notifications/unread-count`. Poll it when the app comes to the foreground and every one or two minutes while open.
- List with `GET /me/notifications` (use `unreadOnly=true` for unread only). Mark as read with `POST /me/notifications/{id}/read` or `POST /me/notifications/read-all`.
- Each notification has **type**, **title**, **body** and **data** (ids such as vehicleId, slug, offerId, enquiryId). Use type and data to open the right screen.
- Users choose channels per type with `GET/PUT /me/notifications/preferences`. Defaults: in-app and email on, SMS off.
- Push notifications (FCM / APNs) are not built yet; see chapter 9.

One notification

```
{
  "id": "01a1124c-...",
  "userId": "01a11242-...",
  "type": "offer.new",
  "title": "You received a new offer",
  "body": "Sandton Auto offers R495 000 for your 2022 BMW 3 Series. Valid until
    2026-10-08T17:30:00.000Z.",
  "data": {
    "offerId": "01a1125a-...",
    "sellRequestId": "01a11250-..."
  },
  "readAt": null,
  "createdAt": "2026-10-06T17:30:00.000Z"
}
```

Type	Sent to	When
<code>enquiry.new</code>	Dealer staff or admins	A new enquiry arrived
<code>enquiry.response</code>	Customer	A dealer or the ChangeCars team replied
<code>lead.assigned</code>	Dealer staff	A lead was assigned to me
<code>valuation.ready</code>	Customer	My vehicle valuation is ready
<code>bidding.opened</code>	Dealer staff	A new vehicle is open for offers
<code>offer.new</code>	Customer	A dealer made or updated an offer
<code>offer.countered</code>	Dealer staff	The customer sent a counter-offer
<code>offer.counter_response</code>	Customer	The dealer accepted or declined my counter-offer
<code>offer.accepted</code>	Customer and dealer	An offer was accepted
<code>offer.rejected</code>	Dealer staff	My offer was rejected, or another offer was accepted
<code>offer.expired</code>	Customer and dealer	An offer expired
<code>offer.withdrawn</code>	Customer	A dealer withdrew an offer
<code>vehicle.status</code>	Dealer staff	A listing was approved, rejected, suspended, released...
<code>vehicle.review_requested</code>	Admins	A listing is waiting for review
<code>saved_search.match</code>	Customer	A new vehicle (or price drop) matches my saved search
<code>favourite.price_drop</code>	Customer	A saved vehicle dropped in price
<code>favourite.status</code>	Customer	A saved vehicle was reserved or sold

Type	Sent to	When
dealer.application	Admins	A new dealer registered
dealer.status	Dealer owner	My dealership was approved, rejected or suspended
dealer.staff_invited	Staff member	I was added to a dealership (email invite)

6.8 Customer account and privacy

- **Profile:** GET /me and PATCH /me (name, phone, marketing consent).
- **Dashboard:** GET /me/dashboard returns counts and the latest enquiries, sell requests and recently viewed vehicles in one call.
- **Devices:** GET /me/sessions lists signed-in devices; DELETE /me/sessions/{id} signs one out.
- **Download my data:** GET /me/export.
- **Delete account:** POST /me/delete-account with the current password. Personal data is anonymised and the user is signed out everywhere.

GET /me/dashboard (shortened)

```
{
  "profile": {
    "id": "01a11252-d4d1-743d-bbbb-eca9ba449aaf",
    "email": "docs.1791308780443@example.com",
    "firstName": "Sipho",
    "lastName": "Dlamini",
    "phone": "+27 82 555 0101",
    "role": "CUSTOMER",
    "marketingConsent": false,
    "createdAt": "2026-10-06T17:46:20.753Z",
    "lastLoginAt": null
  },
  "counts": {
    "openEnquiries": 1,
    "activeOffers": 0,
    "favourites": 1,
    "savedSearches": 0,
    "unreadNotifications": 0
  },
  "latestEnquiries": [
    {
      "id": "01a11252-d6c9-720c-8ab6-34aac7a1153d",
      "reference": "ENQ-CEGAWFE4",
      "type": "VEHICLE",
      "status": "NEW",
      "createdAt": "2026-10-06T17:46:21.257Z",
      "updatedAt": "2026-10-06T17:46:21.257Z",
      "vehicle": {
        "title": "2021 Toyota Hilux 2.4 GD-6 Raider Xtra Cab",
        "slug": "2021-toyota-hilux-2-4-gd-6-raider-xtra-cab-demo3"
      },
      "dealer": {
        "name": "Sandton Auto"
      }
    }
  ],
  "sellRequests": [],
  "recentlyViewed": []
}
```

6.9 Dealer portal

Registration and approval

1. **POST /dealers/register** with an owner object (account details) and a dealership object (business name, contact, province, city, address and optional registration numbers). The response contains tokens, so the owner is signed in at once.
2. The dealership starts as PENDING. Upload verification documents: **POST /dealer/documents/upload-url**, PUT the file, then **POST /dealer/documents**.
3. An admin approves or rejects the application. The owner gets a dealer.status notification.

Branches and staff

- Branches: **GET/POST /dealer/branches**, **PATCH /dealer/branches/{id}** with address, coordinates and opening hours.
- Staff: **POST /dealer/staff** creates the account and emails an invitation. Change role, branch, extra permissions or disable with **PATCH /dealer/staff/{memberId}**. Only the owner can manage managers.

Vehicle lifecycle

A listing moves through fixed states. The server refuses any other move with 409 INVALID_STATUS_TRANSITION and returns the allowed moves in details.

Action	Endpoint (dealer)	From	To	Notes
Create	POST /dealer/vehicles	-	DRAFT	Specs are filled from the catalogue variant
Submit	POST .../{id}/submit	DRAFT, REJECTED	PENDING_REVIEW	Dealership approved; at least one photo
Withdraw	POST .../{id}/withdraw	PENDING_REVIEW	DRAFT	
Approve / reject	Admin only	PENDING_REVIEW	APPROVED / REJECTED	Reject needs a reason
Publish	POST .../{id}/publish	APPROVED	PUBLISHED	Needs INVENTORY_PUBLISH
Reserve	POST .../{id}/reserve	PUBLISHED	RESERVED	Default 72 hours, then released automatically
Release	POST .../{id}/release	RESERVED	PUBLISHED	
Sell	POST .../{id}/sell	PUBLISHED, RESERVED	SOLD	Leaves search results
Suspend	POST .../{id}/suspend	PUBLISHED, RESERVED	SUSPENDED	If an admin suspended it, only an admin can undo it
Unsuspend	POST .../{id}/unsuspend	SUSPENDED	PUBLISHED	
Back to draft	POST .../{id}/edit	REJECTED, APPROVED, PUBLISHED, SUSPENDED	DRAFT	To change make, model, year, mileage...
Archive	POST .../{id}/archive	Most states	ARCHIVED	Removes the listing

What can be edited after approval

In DRAFT and REJECTED everything can be edited. After approval, **PATCH /dealer/vehicles/{id}** only accepts price, specialPrice, isSpecial, description, colour, branchId, stockNumber, featureIds and promotionId. Other fields return 409 MATERIAL_CHANGE_REQUIRES_REVIEW: move the vehicle back to draft first. A lower price on a live vehicle notifies customers who saved it.

Uploading vehicle photos

1. **POST** `/dealer/vehicles/{id}/images/upload-url` with `contentType` (image/jpeg, image/png or image/webp) and `sizeBytes` (max 15 MB). Max 40 photos per vehicle.
2. PUT the file bytes to `uploadUrl` with the returned Content-Type header. The file goes straight to storage, not through the API.
3. **POST** `/dealer/vehicles/{id}/images/{imageId}/complete`. The image status becomes PROCESSING.
4. The worker checks the file is a real image, removes location data and creates three WebP sizes. Status becomes READY (or FAILED if the file is not a valid image). Reload the images to show them.
5. Order photos with **PUT** `.../images/order`; the first photo becomes the main photo.

Enquiries and CRM

- Enquiries: **GET** `/dealer/enquiries`, reply with **POST** `/dealer/enquiries/{id}/respond`. The first reply moves the lead from NEW to CONTACTED and records the response time.
- Leads: **GET** `/dealer/leads` (filters: stage, assignedTo=me|unassigned|memberId, followUpDue=true...). Members without LEADS_VIEW_ALL only see leads assigned to them.
- Stages: **POST** `/dealer/leads/{id}/stage`. Leads move forward only; OFFER_SENT may go back to NEGOTIATION; WON needs QUALIFIED first; LOST needs a lostReason; only managers can reopen closed leads.
- Contact history: **POST** `/dealer/leads/{id}/activities` (call, email, WhatsApp, meeting, note) with an optional next follow-up date.
- Pipeline numbers: **GET** `/dealer/leads/stats`; overall dashboard: **GET** `/dealer/dashboard`.

GET /dealer/leads (one lead shown)

```
{
  "data": [
    {
      "id": "01a11252-d6cd-71f3-9232-0aba3e9c7042",
      "dealerId": "01a11251-673d-7396-8b37-34ce4821d788",
      "branchId": "01a11251-6740-70e9-af43-3e258ad6ec2d",
      "assignedToId": null,
      "customerId": "01a11252-d4d1-743d-bbbb-eca9ba449aaf",
      "enquiryId": "01a11252-d6c9-720c-8ab6-34aac7a1153d",
      "vehicleId": "01a11251-678a-706f-a432-6e6ba34717d4",
      "name": "Sipho Dlamini",
      "email": "docs.1791308780443@example.com",
      "phone": "+27 82 555 0101",
      "source": "WEBSITE_ENQUIRY",
      "stage": "NEW",
      "nextFollowUpAt": null,
      "firstResponseAt": null,
      "lastContactedAt": null,
      "lostReason": null,
      "wonAt": null,
      "lostAt": null,
      "createdAt": "2026-10-06T17:46:21.261Z",
      "updatedAt": "2026-10-06T17:46:21.261Z",
      "vehicle": {
        "id": "01a11251-678a-706f-a432-6e6ba34717d4",
        "title": "2021 Toyota Hilux 2.4 GD-6 Raider Xtra Cab",
        "slug": "2021-toyota-hilux-2-4-gd-6-raider-xtra-cab-demo3",
        "primaryImageUrl": null,
        "price": 409900
      },
      "assignedTo": null,
      "branch": {
        "id": "01a11251-6740-70e9-af43-3e258ad6ec2d",
        "name": "Sandton Auto - Sandton"
      }
    }
  ],
  "meta": {
    "page": 1,
    "pageSize": 1,
    "total": 1,
    "pageCount": 1
  }
}
```

```
}
}
```

Bidding on customer cars

- **GET /dealer/bidding/sessions** lists cars open for offers that this dealership may bid on (approved dealership, bidding enabled, invited or in an eligible province). Customer contact details, registration and VIN are hidden until an offer is accepted.
- Bid with **POST /dealer/bidding/sessions/{id}/offers** (amount, terms, private dealerNotes). Each dealership has one offer per session; posting again revises it (status UPDATED) when the session allows changes. The full history is kept.
- Every offer has an expiry time set by the session. Expired offers can be renewed while bidding is open by posting again.
- Answer a customer counter-offer with **POST /dealer/offers/{id}/counter-response** (accept=true sets your offer to the counter amount).
- When the customer accepts, the dealer gets a notification and a new lead with source OFFER_ACCEPTED.

6.10 Admin panel

Task	Endpoints
Approve dealers	GET /admin/dealers?status=PENDING, GET /admin/dealers/{id} (documents have download links), PATCH /admin/dealers/{id}/status
Review listings	GET /admin/vehicles/review-queue, POST /admin/vehicles/{id}/approve, .../reject
Moderate listings	.../suspend, .../unsuspend, .../archive, .../relist, feature or mark special with PATCH .../flags
Catalogue	/admin/catalogue/*: makes, models, generations, variants with specifications, categories, features and feature assignments
Enquiries	GET /admin/enquiries?platformOnly=true, reply, assign a team member, or route to a dealer (creates a lead)
Sell requests and bidding	Manual valuation POST /admin/sell-requests/{id}/valuations; open bidding POST /admin/sell-requests/{id}/bidding-sessions; reopen or cancel sessions
Content	Articles, media, FAQs, pages, promotions under /admin/...; publish with POST .../{id}/status
Users	GET /admin/users, suspend with PATCH /admin/users/{id}/status; super admins create admins and change roles
Monitoring	GET /admin/stats, GET /admin/audit-logs, GET /admin/system/outbox (background queue health) and retry failed events

Bidding rules are set per session when it is opened: closesAt (required), opensAt, allowBidModification, allowBidWithdrawal, allowCounterOffers, offerValidityHours (default 48), reservePrice, eligibleProvinces and inviteDealerIds. A vehicle can only have one live bidding session. A closed session accepts no new bids unless it is explicitly reopened.

6.11 Content and SEO

Articles and static pages store their text as a list of **blocks**, never raw HTML. This keeps the apps safe from script injection. Render each block with native components:

Block	Fields	Render as
paragraph	text	Normal text
heading	text, level (2, 3 or 4)	Heading
image	src (https), alt	Image with alt text
link	text, href (https or /relative)	Link or button
list	items (array of text)	Bullet list
quote	text	Quote
video	url (https)	Embedded video

- Articles: `GET /content/articles` (filters q, category, type, tag), `GET /content/articles/{slug}`.
- Videos and podcasts: `GET /content/media?type=VIDEO`. FAQs: `GET /content/faqs?category=selling`.
- Static pages by slug: `privacy-policy`, `terms-and-conditions`, `insurance`, `electric-vehicles`, `ev-charging` via `GET /content/pages/{slug}`.
- Promotions: `GET /promotions` and `GET /promotions/{slug}` with their vehicles.
- Sitemaps for search engines: `GET /seo/sitemap.xml` (index). Each article and page has metaTitle and metaDescription for SEO tags.

7. Status and value reference

These are the fixed values (enums) used in requests and responses. Use the value on the left in API calls and show your own label in the app.

Vehicle status (lifecycle)

Value	Meaning
DRAFT	Being prepared by the dealer. Not visible to the public.
PENDING_REVIEW	Submitted. Waiting for an admin to check it.
APPROVED	Checked by an admin. The dealer can now publish it.
REJECTED	Not approved. reviewNotes explains why. The dealer can fix and resubmit.
PUBLISHED	Live on the website and in search results.
RESERVED	Still visible, but marked as reserved for a buyer. Expires automatically.
SOLD	Sold. Removed from search. Detail page still works for old links.
SUSPENDED	Temporarily offline (by the dealer or an admin).
ARCHIVED	Removed. Kept only for history.

Vehicle availability (what to show the customer)

Value	Meaning
AVAILABLE	Show 'Available'. Enquiries allowed.
RESERVED	Show a 'Reserved' badge. Enquiries still allowed (backup buyers).
SOLD	Show 'Sold'. Hide enquiry and finance buttons.
TEMPORARILY_UNAVAILABLE	Listing is suspended.
ARCHIVED / NOT_LISTED	Not for sale. Only seen in dealer and admin tools.

Vehicle condition

Value	Meaning
NEW	New vehicle
USED	Pre-owned vehicle
DEMO	Dealer demonstrator

Fuel type

Value	Meaning
PETROL	Petrol
DIESEL	Diesel
HYBRID	Hybrid
PLUGIN_HYBRID	Plug-in hybrid
ELECTRIC	Electric
LPG	LPG
OTHER	Other

Transmission

Value	Meaning
MANUAL	Manual
AUTOMATIC	Automatic

Drivetrain

Value	Meaning
FWD	Front-wheel drive
RWD	Rear-wheel drive
AWD	All-wheel drive
FOUR_X_TWO	4x2
FOUR_X_FOUR	4x4

Province

Value	Meaning
EASTERN_CAPE	Eastern Cape
FREE_STATE	Free State
GAUTENG	Gauteng
KWAZULU_NATAL	KwaZulu-Natal
LIMPOPO	Limpopo
MPUMALANGA	Mpumalanga
NORTHERN_CAPE	Northern Cape
NORTH_WEST	North West
WESTERN_CAPE	Western Cape

Enquiry type

Value	Meaning
VEHICLE	Contact dealer about a listed vehicle
TEST_DRIVE	Test drive booking
QUOTE	Quote request for a (new) vehicle
BEAT_MY_QUOTE	Beat My Quote request
TRADE_IN	Trade-in request
CONCIERGE	Concierge service request
HELP_ME_FIND	Help me find a vehicle
FINANCE	Finance enquiry
INSURANCE	Insurance quote request
GENERAL	Contact us message

Enquiry status

Value	Meaning
NEW	Received, nobody replied yet
IN_PROGRESS	Being handled, or the customer sent a follow-up
RESPONDED	The dealer or team replied
CLOSED	Finished
CANCELLED	Cancelled by the customer

Lead stage (dealer CRM)

Value	Meaning
NEW	New lead, not contacted yet
CONTACTED	Dealer contacted the customer (set automatically on first reply)
QUALIFIED	Serious buyer with a real need and budget
NEGOTIATION	Discussing price and terms
OFFER_SENT	A formal offer was sent
WON	Deal done
LOST	Lost. lostReason is required

Lead source

Value	Meaning
WEBSITE_ENQUIRY	Vehicle, test drive, finance or insurance enquiry
QUOTE_REQUEST	Quote or Beat My Quote
TRADE_IN	Trade-in request
OFFER_ACCEPTED	Customer accepted this dealer's offer on their car
PHONE / WHATSAPP / EMAIL / WALK_IN / OTHER	Created manually by the dealer

Sell request type

Value	Meaning
SELL	Customer wants dealer offers
VALUATION	Customer only wants to know the value

Sell request status

Value	Meaning
SUBMITTED	Received. Automatic valuation is running
UNDER_REVIEW	Waiting for the ChangeCars team (manual valuation or offers requested)
VALUED	A valuation is available
BIDDING	Dealers are making offers
OFFER_ACCEPTED	The customer accepted an offer. A deal exists
COMPLETED	The sale is finished
CANCELLED	Cancelled by the customer or team
REJECTED	Not accepted by the team

Vehicle condition grade (sell requests)

Value	Meaning
EXCELLENT	Like new
GOOD	Normal wear
FAIR	Visible wear or small issues
POOR	Needs work

Bidding session status

Value	Meaning
SCHEDULED	Opens in the future
OPEN	Dealers can bid now
CLOSED	Bidding time ended. Existing offers can still be accepted until they expire
AWARDED	An offer was accepted
CANCELLED	Cancelled

Offer status

Value	Meaning
SUBMITTED	New offer from a dealer
UPDATED	The dealer changed the amount or terms
PENDING	The customer sent a counter-offer. Waiting for the dealer
ACCEPTED	The customer accepted it
REJECTED	The customer rejected it (or bidding was cancelled)
EXPIRED	Its validity time ended
WITHDRAWN	The dealer withdrew it
SUPERSEDED	Another offer was accepted

Counter-offer status

Value	Meaning
OPEN	Waiting for the dealer
ACCEPTED	Dealer accepted; offer amount updated
DECLINED	Dealer declined
CANCELLED	No longer relevant

Dealer status

Value	Meaning
PENDING	Registered, waiting for admin approval
APPROVED	Can publish listings and bid
REJECTED	Application rejected
SUSPENDED	Blocked by an admin

Dealer member role

Value	Meaning
OWNER	Dealership owner. All permissions
MANAGER	Manages inventory, staff and leads
SALES	Handles enquiries and assigned leads
STAFF	Basic access to assigned leads

Platform user role

Value	Meaning
CUSTOMER	Normal user
DEALER	Belongs to a dealership
ADMIN	ChangeCars team member
SUPER_ADMIN	Full control, manages admins

Content status

Value	Meaning
DRAFT	Not public
PUBLISHED	Public (from publishedAt)
ARCHIVED	Hidden

Article type

Value	Meaning
NEWS	News
REVIEW	Vehicle review
ADVICE	Car-buying advice
GUIDE	Guide
ARTICLE	General article

Media type

Value	Meaning
VIDEO	Video
PODCAST	Podcast

8. API reference

This chapter lists every endpoint, generated from the running backend so it matches the code exactly. Paths are shown in full. In a path, **{id}** means you replace it with a value.

Label	Meaning
Public (no login)	No token needed.
Public. Login optional	Works without a token; with a token the request is linked to the user.
Signed-in user (any role)	Send a valid access token.
Dealer member, permission X	The user must be an active member of a dealership with permission X (see 5.6).
Admin / Super admin only	The user must have the ADMIN or SUPER_ADMIN role.
Success	HTTP status code on success. 204 means no response body.

Request body tables list every field. Nested fields are written with dots, for example owner.email, and list items with [], for example items[].channel. Response bodies follow the shapes shown in chapters 4 to 6 and appendix B. Swagger UI (/docs) lets you try any endpoint.

8.1 Public and customer APIs

Used by the customer mobile app and the public website. Endpoints marked Public need no login.

Health · 2 endpoints

Probes for load balancers and uptime monitors. Not under /api/v1.

GET /health/live

Liveness: the process is up.

Access: Public (no login). **Success:** 200.

GET /health/ready

Readiness: dependencies reachable (returns 503 otherwise).

Access: Public (no login). **Success:** 200.

Auth · 9 endpoints

Sign up, sign in, tokens and passwords. See chapter 5 for the full token flow.

POST /api/v1/auth/change-password

Change password and sign out other devices.

Access: Signed-in user (any role). **Success:** 204.

Body field	Type	Required	Rules and notes
currentPassword	string	Yes	min length 1; max length 128
newPassword	string	Yes	

POST /api/v1/auth/forgot-password

Email a password reset link (always returns 202).

Access: Public (no login). **Success:** 202. Rate limit 5 requests per minute per IP.

Body field	Type	Required	Rules and notes
email	string (email)	Yes	

POST /api/v1/auth/login

Sign in with email and password.

Access: Public (no login). **Success:** 200. Rate limit 10 requests per minute per IP.

Body field	Type	Required	Rules and notes
email	string (email)	Yes	
password	string	Yes	min length 1; max length 128

POST /api/v1/auth/logout

Revoke the current session.

Access: Signed-in user (any role). **Success:** 204.

POST /api/v1/auth/logout-all

Revoke every session of the current user.

Access: Signed-in user (any role). **Success:** 204.

GET /api/v1/auth/me

Current user with dealer membership, if any.

Access: Signed-in user (any role). **Success:** 200.

POST /api/v1/auth/refresh

Exchange a refresh token for new tokens (rotation).

Access: Public (no login). **Success:** 200. Rate limit 30 requests per minute per IP.

Body field	Type	Required	Rules and notes
refreshToken	string	Yes	min length 20; max length 200

POST /api/v1/auth/register

Create a customer account.

Access: Public (no login). **Success:** 201. Rate limit 10 requests per minute per IP.

Body field	Type	Required	Rules and notes
email	string (email)	Yes	max length 254. e.g. "thabo@example.com"
password	string	Yes	e.g. "Sup3rSecret!"
firstName	string	Yes	min length 1; max length 80. e.g. "Thabo"
lastName	string	Yes	min length 1; max length 80. e.g. "Nkosi"
phone	string	No	e.g. "+27 82 123 4567"
marketingConsent	boolean	No	default false
acceptTerms	boolean	Yes	Must be true: user accepts terms and privacy policy (NFR-17)

POST /api/v1/auth/reset-password

Set a new password with a reset token.

Access: Public (no login). **Success:** 204. Rate limit 10 requests per minute per IP.

Body field	Type	Required	Rules and notes
token	string	Yes	min length 20; max length 200
password	string	Yes	

Catalogue · 7 endpoints

Makes, models, variants, specifications, categories and features. Changes rarely, so cache it in the app.

GET `/api/v1/catalogue/categories`

Vehicle categories: hatchbacks, SUVs, bakkies, EVs...

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 3600 s.

GET `/api/v1/catalogue/features`

All vehicle features (safety, comfort, technology...).

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 3600 s.

GET `/api/v1/catalogue/makes`

All active manufacturers.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 3600 s.

GET `/api/v1/catalogue/makes/{makeSlug}/models`

Models of a manufacturer.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 3600 s.

Parameter	In	Type	Required	Notes
<code>makeSlug</code>	path	string	Yes	

GET `/api/v1/catalogue/makes/{makeSlug}/models/{modelSlug}`

Model with generations, variants, specs and pricing.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 3600 s.

Parameter	In	Type	Required	Notes
<code>makeSlug</code>	path	string	Yes	
<code>modelSlug</code>	path	string	Yes	

GET `/api/v1/catalogue/variants/compare`

Compare 2-4 catalogue variants side by side.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 3600 s.

Parameter	In	Type	Required	Notes
<code>ids</code>	query	string	Yes	Comma-separated ids (2-4). max length 200. e.g. " <code>id1,id2</code> "

GET `/api/v1/catalogue/variants/{id}`

Variant with full specification and resolved features.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 3600 s.

Parameter	In	Type	Required	Notes
<code>id</code>	path	string	Yes	

Vehicles (public) · 9 endpoints

Search, filters, vehicle pages, compare and related vehicles.

GET

/api/v1/vehicles

Search, filter and sort listed vehicles.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 60 s.

Parameter	In	Type	Required	Notes
q	query	string	No	Free text over the listing title. max length 100
make	query	string	No	Make slugs, comma-separated. max length 500. e.g. "bmw,audi"
model	query	string	No	Model slugs; scope to a make with make:model. max length 500. e.g. "bmw:3-series"
variant	query	string	No	Variant slugs. max length 500. e.g. "320i-m-sport-2022"
condition	query	string	No	NEW, USED, DEMO. max length 50. e.g. "USED"
category	query	string	No	Category slugs (FR-25). max length 300. e.g. "suvs,bakkies"
fuelType	query	string	No	max length 200. e.g. "PETROL,DIESEL"
transmission	query	string	No	max length 50. e.g. "AUTOMATIC"
drivetrain	query	string	No	max length 100. e.g. "FOUR_X_FOUR"
province	query	string	No	Provinces (FR-27). max length 300. e.g. "GAUTENG,WESTERN_CAPE"
city	query	string	No	max length 100
colour	query	string	No	max length 200. e.g. "White,Black"
dealer	query	string	No	Dealer slug. max length 120
branchId	query	string (uuid)	No	
minPrice	query	number	No	min 0
maxPrice	query	number	No	min 0
minYear	query	number	No	min 1900
maxYear	query	number	No	max 2100
minMileage	query	number	No	min 0
maxMileage	query	number	No	min 0
minEngineCc	query	number	No	min 0
maxEngineCc	query	number	No	min 0
minPowerKw	query	number	No	min 0
maxPowerKw	query	number	No	min 0
seats	query	string	No	Seat counts, comma-separated; "8+" for eight or more. max length 50. e.g. "5,7"
onSpecial	query	boolean	No	Only vehicles on special (FR-23)
featured	query	boolean	No	
availableOnly	query	boolean	No	Hide reserved vehicles
collection	query	enum	No	One of: hot-sellers, specials, featured, new-arrivals, budget, student, bakkies, electric
lat	query	number	No	Centre latitude for radius search
lng	query	number	No	Centre longitude for radius search
radiusKm	query	number	No	Radius in km (with lat/lng). min 1; max 1000
sort	query	enum	No	One of: recent, oldest, price-asc, price-desc, mileage-asc, mileage-desc, year-desc, year-asc, popular; default "recent"

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; max 500; default 1
pageSize	query	number	No	min 1; max 50; default 20

GET /api/v1/vehicles/compare

Compare 2-4 vehicles: price, engine, performance, consumption, dimensions, features, warranty.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 60 s.

Parameter	In	Type	Required	Notes
ids	query	string	Yes	Comma-separated vehicle ids (2-4). max length 200

GET /api/v1/vehicles/facets

Counts per make, fuel, transmission, province, condition and category for a search.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 60 s.

Parameter	In	Type	Required	Notes
q	query	string	No	Free text over the listing title. max length 100
make	query	string	No	Make slugs, comma-separated. max length 500. e.g. "bmw,audi"
model	query	string	No	Model slugs; scope to a make with make:model. max length 500. e.g. "bmw:3-series"
variant	query	string	No	Variant slugs. max length 500. e.g. "320i-m-sport-2022"
condition	query	string	No	NEW, USED, DEMO. max length 50. e.g. "USED"
category	query	string	No	Category slugs (FR-25). max length 300. e.g. "suvs,bakkies"
fuelType	query	string	No	max length 200. e.g. "PETROL,DIESEL"
transmission	query	string	No	max length 50. e.g. "AUTOMATIC"
drivetrain	query	string	No	max length 100. e.g. "FOUR_X_FOUR"
province	query	string	No	Provinces (FR-27). max length 300. e.g. "GAUTENG,WESTERN_CAPE"
city	query	string	No	max length 100
colour	query	string	No	max length 200. e.g. "White,Black"
dealer	query	string	No	Dealer slug. max length 120
branchId	query	string (uuid)	No	
minPrice	query	number	No	min 0
maxPrice	query	number	No	min 0
minYear	query	number	No	min 1900
maxYear	query	number	No	max 2100
minMileage	query	number	No	min 0
maxMileage	query	number	No	min 0
minEngineCc	query	number	No	min 0
maxEngineCc	query	number	No	min 0
minPowerKw	query	number	No	min 0
maxPowerKw	query	number	No	min 0
seats	query	string	No	Seat counts, comma-separated; "8+" for eight or more. max length 50. e.g. "5,7"
onSpecial	query	boolean	No	Only vehicles on special (FR-23)

Parameter	In	Type	Required	Notes
featured	query	boolean	No	
availableOnly	query	boolean	No	Hide reserved vehicles
collection	query	enum	No	One of: hot-sellers , specials , featured , new-arrivals , budget , student , bakkies , electric
lat	query	number	No	Centre latitude for radius search
lng	query	number	No	Centre longitude for radius search
radiusKm	query	number	No	Radius in km (with lat/lng). min 1; max 1000

GET /api/v1/vehicles/hot-sellers

Popular, high-demand vehicles.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 60 s.

Parameter	In	Type	Required	Notes
q	query	string	No	Free text over the listing title. max length 100
make	query	string	No	Make slugs, comma-separated. max length 500. e.g. "bmw,audi"
model	query	string	No	Model slugs; scope to a make with make:model. max length 500. e.g. "bmw:3-series"
variant	query	string	No	Variant slugs. max length 500. e.g. "320i-m-sport-2022"
condition	query	string	No	NEW, USED, DEMO. max length 50. e.g. "USED"
category	query	string	No	Category slugs (FR-25). max length 300. e.g. "suvs,bakkies"
fuelType	query	string	No	max length 200. e.g. "PETROL,DIESEL"
transmission	query	string	No	max length 50. e.g. "AUTOMATIC"
drivetrain	query	string	No	max length 100. e.g. "FOUR_X_FOUR"
province	query	string	No	Provinces (FR-27). max length 300. e.g. "GAUTENG,WESTERN_CAPE"
city	query	string	No	max length 100
colour	query	string	No	max length 200. e.g. "White,Black"
dealer	query	string	No	Dealer slug. max length 120
branchId	query	string (uuid)	No	
minPrice	query	number	No	min 0
maxPrice	query	number	No	min 0
minYear	query	number	No	min 1900
maxYear	query	number	No	max 2100
minMileage	query	number	No	min 0
maxMileage	query	number	No	min 0
minEngineCc	query	number	No	min 0
maxEngineCc	query	number	No	min 0
minPowerKw	query	number	No	min 0
maxPowerKw	query	number	No	min 0
seats	query	string	No	Seat counts, comma-separated; "8+" for eight or more. max length 50. e.g. "5,7"
onSpecial	query	boolean	No	Only vehicles on special (FR-23)
featured	query	boolean	No	
availableOnly	query	boolean	No	Hide reserved vehicles

Parameter	In	Type	Required	Notes
collection	query	enum	No	One of: hot-sellers , specials , featured , new-arrivals , budget , student , bakkies , electric
lat	query	number	No	Centre latitude for radius search
lng	query	number	No	Centre longitude for radius search
radiusKm	query	number	No	Radius in km (with lat/lng). min 1; max 1000
sort	query	enum	No	One of: recent , oldest , price-asc , price-desc , mileage-asc , mileage-desc , year-desc , year-asc , popular ; default "recent"
page	query	number	No	min 1; max 500; default 1
pageSize	query	number	No	min 1; max 50; default 20

GET /api/v1/vehicles/specials

Vehicles on special.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 60 s.

Parameter	In	Type	Required	Notes
q	query	string	No	Free text over the listing title. max length 100
make	query	string	No	Make slugs, comma-separated. max length 500. e.g. "bmw,audi"
model	query	string	No	Model slugs; scope to a make with make:model. max length 500. e.g. "bmw:3-series"
variant	query	string	No	Variant slugs. max length 500. e.g. "320i-m-sport-2022"
condition	query	string	No	NEW, USED, DEMO. max length 50. e.g. "USED"
category	query	string	No	Category slugs (FR-25). max length 300. e.g. "suvs,bakkies"
fuelType	query	string	No	max length 200. e.g. "PETROL,DIESEL"
transmission	query	string	No	max length 50. e.g. "AUTOMATIC"
drivetrain	query	string	No	max length 100. e.g. "FOUR_X_FOUR"
province	query	string	No	Provinces (FR-27). max length 300. e.g. "GAUTENG,WESTERN_CAPE"
city	query	string	No	max length 100
colour	query	string	No	max length 200. e.g. "White,Black"
dealer	query	string	No	Dealer slug. max length 120
branchId	query	string (uuid)	No	
minPrice	query	number	No	min 0
maxPrice	query	number	No	min 0
minYear	query	number	No	min 1900
maxYear	query	number	No	max 2100
minMileage	query	number	No	min 0
maxMileage	query	number	No	min 0
minEngineCc	query	number	No	min 0
maxEngineCc	query	number	No	min 0
minPowerKw	query	number	No	min 0
maxPowerKw	query	number	No	min 0
seats	query	string	No	Seat counts, comma-separated; "8+" for eight or more. max length 50. e.g. "5,7"

Parameter	In	Type	Required	Notes
onSpecial	query	boolean	No	Only vehicles on special (FR-23)
featured	query	boolean	No	
availableOnly	query	boolean	No	Hide reserved vehicles
collection	query	enum	No	One of: hot-sellers, specials, featured, new-arrivals, budget, student, bakkies, electric
lat	query	number	No	Centre latitude for radius search
lng	query	number	No	Centre longitude for radius search
radiusKm	query	number	No	Radius in km (with lat/lng). min 1; max 1000
sort	query	enum	No	One of: recent, oldest, price-asc, price-desc, mileage-asc, mileage-desc, year-desc, year-asc, popular; default "recent"
page	query	number	No	min 1; max 500; default 1
pageSize	query	number	No	min 1; max 50; default 20

GET /api/v1/vehicles/{id}/dealer-vehicles

Other vehicles from the same dealer.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 60 s.

Parameter	In	Type	Required	Notes
id	path	string	Yes	
limit	query	number	No	min 1; max 24; default 6

GET /api/v1/vehicles/{id}/market-price

Average asking price of comparable listings.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 60 s.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

GET /api/v1/vehicles/{id}/similar

Similar vehicles (same model first, then same body type at a similar price).

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 60 s.

Parameter	In	Type	Required	Notes
id	path	string	Yes	
limit	query	number	No	min 1; max 24; default 6

GET /api/v1/vehicles/{slugOrId}

Vehicle detail page. Signed-in viewers get it added to recently viewed.

Access: Public. Login optional (when signed in, the request is linked to the user). **Success:** 200. Anonymous responses are CDN-cacheable for 60 s.

Parameter	In	Type	Required	Notes
slugOrId	path	string	Yes	

Finance · 2 endpoints

Finance and affordability calculators. No login needed.

POST `/api/v1/finance/affordability`

"What can I afford" vehicle budget estimate.

Access: Public (no login). **Success:** 200.

Body field	Type	Required	Rules and notes
monthlyIncome	number	Yes	e.g. 45000
monthlyExpenses	number	Yes	e.g. 25000
deposit	number	Yes	e.g. 30000
termMonths	number	Yes	e.g. 72
annualInterestRate	number	Yes	e.g. 11.75
balloonPercent	number	No	e.g. 0

POST `/api/v1/finance/repayment`

Monthly repayment, finance amount and total repayment.

Access: Public (no login). **Success:** 200.

Body field	Type	Required	Rules and notes
vehiclePrice	number	Yes	e.g. 450000
deposit	number	Yes	e.g. 45000
termMonths	number	Yes	default 72. e.g. 72
annualInterestRate	number	Yes	e.g. 11.75
balloonPercent	number	No	e.g. 30

Enquiries (public forms) · 10 endpoints

All customer request forms. Login is optional; when the user is signed in the enquiry appears in their dashboard.

POST `/api/v1/enquiries/beat-my-quote`

Beat My Quote: submit an existing quote for a better offer.

Access: Public. Login optional (when signed in, the request is linked to the user). **Success:** 201. Rate limit 10 requests per minute per IP.

Body field	Type	Required	Rules and notes
name	string	Yes	min length 2; max length 160. e.g. "Thabo Nkosi"
email	string (email)	Yes	max length 254
phone	string	Yes	e.g. "+27 82 123 4567"
consent	boolean	Yes	Consent to be contacted about this request (POPIA / NFR-17)
makeId	string (uuid)	No	
modelId	string (uuid)	No	
variantId	string (uuid)	No	
desiredVehicle	string	Yes	Desired vehicle as quoted. min length 3; max length 200. e.g. "2025 Toyota Hilux 2.8 GD-6 Legend"
quotedPrice	number	Yes	Quoted price in rand. min 1000; max 100000000
quotingDealer	string	Yes	Dealer that issued the existing quote. min length 2; max length 200
quoteDetails	string	No	Quote details / reference / included extras. max length 4000

Body field	Type	Required	Rules and notes
province	enum	Yes	One of: EASTERN_CAPE, FREE_STATE, GAUTENG, KWAZULU_NATAL, LIMPOPO, MPUMALANGA, NORTHERN_CAPE, NORTH_WEST, WESTERN_CAPE

POST /api/v1/enquiries/concierge

Request concierge services.

Access: Public. Login optional (when signed in, the request is linked to the user). **Success:** 201. Rate limit 10 requests per minute per IP.

Body field	Type	Required	Rules and notes
name	string	Yes	min length 2; max length 160. e.g. "Thabo Nkosi"
email	string (email)	Yes	max length 254
phone	string	Yes	e.g. "+27 82 123 4567"
consent	boolean	Yes	Consent to be contacted about this request (POPIA / NFR-17)
services	array of enum	Yes	One of: VEHICLE_SOURCING, RECOMMENDATIONS, PRICE_NEGOTIATION, VEHICLE_CHECKS, FINANCE_ASSISTANCE, WARRANTY_ASSISTANCE, INSURANCE_ASSISTANCE, DOCUMENTATION_ASSISTANCE, DELIVERY_COORDINATION; min items 1
budget	number	No	min 0
province	enum	No	One of: EASTERN_CAPE, FREE_STATE, GAUTENG, KWAZULU_NATAL, LIMPOPO, MPUMALANGA, NORTHERN_CAPE, NORTH_WEST, WESTERN_CAPE
notes	string	No	max length 4000

POST /api/v1/enquiries/contact

General "contact us" message.

Access: Public. Login optional (when signed in, the request is linked to the user). **Success:** 201. Rate limit 10 requests per minute per IP.

Body field	Type	Required	Rules and notes
name	string	Yes	min length 2; max length 160. e.g. "Thabo Nkosi"
email	string (email)	Yes	max length 254
phone	string	Yes	e.g. "+27 82 123 4567"
consent	boolean	Yes	Consent to be contacted about this request (POPIA / NFR-17)
subject	string	Yes	min length 2; max length 200
message	string	Yes	min length 2; max length 4000

POST /api/v1/enquiries/finance

Finance application interest.

Access: Public. Login optional (when signed in, the request is linked to the user). **Success:** 201. Rate limit 10 requests per minute per IP.

Body field	Type	Required	Rules and notes
name	string	Yes	min length 2; max length 160. e.g. "Thabo Nkosi"
email	string (email)	Yes	max length 254
phone	string	Yes	e.g. "+27 82 123 4567"
consent	boolean	Yes	Consent to be contacted about this request (POPIA / NFR-17)
vehicleId	string (uuid)	No	
vehiclePrice	number	No	min 0

Body field	Type	Required	Rules and notes
deposit	number	No	min 0
termMonths	number	No	min 6; max 96
monthlyIncome	number	No	min 0
employmentStatus	enum	No	One of: EMPLOYED, SELF_EMPLOYED, CONTRACT, PENSIONER, OTHER
message	string	No	max length 2000

POST /api/v1/enquiries/help-me-find

Ask the team to find a vehicle.

Access: Public. Login optional (when signed in, the request is linked to the user). **Success:** 201. Rate limit 10 requests per minute per IP.

Body field	Type	Required	Rules and notes
name	string	Yes	min length 2; max length 160. e.g. "Thabo Nkosi"
email	string (email)	Yes	max length 254
phone	string	Yes	e.g. "+27 82 123 4567"
consent	boolean	Yes	Consent to be contacted about this request (POPIA / NFR-17)
makeId	string (uuid)	No	
modelId	string (uuid)	No	
vehicleType	string	No	Category slug, e.g. suvs. max length 80
budgetMin	number	No	min 0
budgetMax	number	Yes	min 1000
minYear	number	No	min 1950
maxMileage	number	No	min 0
province	enum	No	One of: EASTERN_CAPE, FREE_STATE, GAUTENG, KWAZULU_NATAL, LIMPOPO, MPUMALANGA, NORTHERN_CAPE, NORTH_WEST, WESTERN_CAPE
preferences	string	No	Colour, transmission, fuel, must-have features.... max length 4000

POST /api/v1/enquiries/insurance

Insurance quote request.

Access: Public. Login optional (when signed in, the request is linked to the user). **Success:** 201. Rate limit 10 requests per minute per IP.

Body field	Type	Required	Rules and notes
name	string	Yes	min length 2; max length 160. e.g. "Thabo Nkosi"
email	string (email)	Yes	max length 254
phone	string	Yes	e.g. "+27 82 123 4567"
consent	boolean	Yes	Consent to be contacted about this request (POPIA / NFR-17)
vehicleId	string (uuid)	No	
vehicleDescription	string	No	max length 200. e.g. "2022 Toyota Corolla Cross 1.8 Xi"
coverType	enum	No	One of: COMPREHENSIVE, THIRD_PARTY_FIRE_THEFT, THIRD_PARTY, NOT_SURE
province	enum	No	One of: EASTERN_CAPE, FREE_STATE, GAUTENG, KWAZULU_NATAL, LIMPOPO, MPUMALANGA, NORTHERN_CAPE, NORTH_WEST, WESTERN_CAPE
message	string	No	max length 2000

POST /api/v1/enquiries/quote

Request a quotation for a vehicle.

Access: Public. Login optional (when signed in, the request is linked to the user). **Success:** 201. Rate limit 10 requests per minute per IP.

Body field	Type	Required	Rules and notes
name	string	Yes	min length 2; max length 160. e.g. "Thabo Nkosi"
email	string (email)	Yes	max length 254
phone	string	Yes	e.g. "+27 82 123 4567"
consent	boolean	Yes	Consent to be contacted about this request (POPIA / NFR-17)
makeId	string (uuid)	Yes	
modelId	string (uuid)	Yes	
variantId	string (uuid)	No	
dealerId	string (uuid)	No	Send to a specific dealer; otherwise the platform team handles it
province	enum	Yes	One of: EASTERN_CAPE, FREE_STATE, GAUTENG, KWAZULU_NATAL, LIMPOPO, MPUMALANGA, NORTHERN_CAPE, NORTH_WEST, WESTERN_CAPE
requirements	string	No	Colour, extras, timing, finance needs.... max length 4000

POST /api/v1/enquiries/test-drive

Book a test drive for a listed vehicle.

Access: Public. Login optional (when signed in, the request is linked to the user). **Success:** 201. Rate limit 10 requests per minute per IP.

Body field	Type	Required	Rules and notes
name	string	Yes	min length 2; max length 160. e.g. "Thabo Nkosi"
email	string (email)	Yes	max length 254
phone	string	Yes	e.g. "+27 82 123 4567"
consent	boolean	Yes	Consent to be contacted about this request (POPIA / NFR-17)
vehicleId	string (uuid)	Yes	
preferredAt	string	Yes	Preferred date/time (ISO 8601)
message	string	No	max length 2000

POST /api/v1/enquiries/trade-in

Trade in a vehicle when buying another.

Access: Public. Login optional (when signed in, the request is linked to the user). **Success:** 201. Rate limit 10 requests per minute per IP.

Body field	Type	Required	Rules and notes
name	string	Yes	min length 2; max length 160. e.g. "Thabo Nkosi"
email	string (email)	Yes	max length 254
phone	string	Yes	e.g. "+27 82 123 4567"
consent	boolean	Yes	Consent to be contacted about this request (POPIA / NFR-17)
vehicleId	string (uuid)	No	Listed vehicle the customer wants (routes to its dealer)
desiredVehicle	string	No	Desired replacement vehicle if not a specific listing. max length 200
tradeIn	object	Yes	
tradeIn.make	string	Yes	min length 1; max length 80. e.g. "Volkswagen"
tradeIn.model	string	Yes	min length 1; max length 80. e.g. "Polo"

Body field	Type	Required	Rules and notes
tradeIn.variant	string	No	max length 120. e.g. "1.0 TSI Comfortline"
tradeIn.year	number	Yes	min 1950; max 2100
tradeIn.mileage	number	Yes	min 0; max 2000000
tradeIn.condition	enum	Yes	One of: EXCELLENT, GOOD, FAIR, POOR
tradeIn.hasServiceHistory	boolean	No	
tradeIn.hasOutstandingFinance	boolean	No	
message	string	No	max length 2000

POST /api/v1/enquiries/vehicle

Contact the dealer about a vehicle. Opens a lead for the dealer.

Access: Public. Login optional (when signed in, the request is linked to the user). **Success:** 201. Rate limit 10 requests per minute per IP.

Body field	Type	Required	Rules and notes
name	string	Yes	min length 2; max length 160. e.g. "Thabo Nkosi"
email	string (email)	Yes	max length 254
phone	string	Yes	e.g. "+27 82 123 4567"
consent	boolean	Yes	Consent to be contacted about this request (POPIA / NFR-17)
vehicleId	string (uuid)	Yes	
message	string	No	max length 4000
interestedInFinance	boolean	No	Customer would like finance
hasTradeIn	boolean	No	Customer has a vehicle to trade in

Sell / value your vehicle · 1 endpoint

Start a valuation or a sell request.

POST /api/v1/sell-requests

Submit your vehicle for a valuation or to receive dealer offers.

Access: Public. Login optional (when signed in, the request is linked to the user). **Success:** 201. Rate limit 10 requests per minute per IP.

Body field	Type	Required	Rules and notes
name	string	Yes	min length 2; max length 160. e.g. "Thabo Nkosi"
email	string (email)	Yes	max length 254
phone	string	Yes	e.g. "+27 82 123 4567"
consent	boolean	Yes	Consent to be contacted about this request (POPIA / NFR-17)
type	enum	Yes	SELL = wants offers, VALUATION = estimate only. One of: SELL, VALUATION
province	enum	Yes	One of: EASTERN_CAPE, FREE_STATE, GAUTENG, KWAZULU_NATAL, LIMPOPO, MPUMALANGA, NORTHERN_CAPE, NORTH_WEST, WESTERN_CAPE
city	string	No	max length 120
makeId	string (uuid)	No	Catalogue ids give the most accurate valuation
modelId	string (uuid)	No	
variantId	string (uuid)	No	
makeName	string	No	Required when makeld is not given. min length 1; max length 80

Body field	Type	Required	Rules and notes
modelName	string	No	Required when modelId is not given. min length 1; max length 80
variantName	string	No	max length 120
year	number	Yes	min 1950; max 2027
mileage	number	Yes	min 0; max 2000000
condition	enum	Yes	One of: EXCELLENT, GOOD, FAIR, POOR
colour	string	No	max length 40
transmission	enum	No	One of: MANUAL, AUTOMATIC
fuelType	enum	No	One of: PETROL, DIESEL, HYBRID, PLUGIN_HYBRID, ELECTRIC, LPG, OTHER
registrationNumber	string	No	Registration number (FR-09). Never shown to dealers before acceptance.. max length 20
vin	string	No	
hasServiceHistory	boolean	No	
hasAccidentHistory	boolean	No	
hasOutstandingFinance	boolean	No	
askingPrice	number	No	Price the customer hopes for. min 0
notes	string	No	max length 4000

Dealers (public) · 3 endpoints

Dealer directory, dealer pages and dealer registration.

GET /api/v1/dealers

Directory of approved dealers.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 300 s.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
q	query	string	No	max length 100
province	query	enum	No	One of: EASTERN_CAPE, FREE_STATE, GAUTENG, KWAZULU_NATAL, LIMPOPO, MPUMALANGA, NORTHERN_CAPE, NORTH_WEST, WESTERN_CAPE

POST /api/v1/dealers/register

Register a dealership. Account starts PENDING until an admin approves it.

Access: Public (no login). **Success:** 201. Rate limit 5 requests per minute per IP. Anonymous responses are CDN-cacheable for 300 s.

Body field	Type	Required	Rules and notes
owner	object	Yes	Owner login credentials and contact person
owner.email	string (email)	Yes	max length 254. e.g. "thabo@example.com"
owner.password	string	Yes	e.g. "Sup3rSecret!"
owner.firstName	string	Yes	min length 1; max length 80. e.g. "Thabo"
owner.lastName	string	Yes	min length 1; max length 80. e.g. "Nkosi"
owner.phone	string	No	e.g. "+27 82 123 4567"
owner.marketingConsent	boolean	No	default false

Body field	Type	Required	Rules and notes
owner.acceptTerms	boolean	Yes	Must be true: user accepts terms and privacy policy (NFR-17)
dealership	object	Yes	
dealership.name	string	Yes	min length 2; max length 120. e.g. "Sandton Auto"
dealership.legalName	string	No	max length 200
dealership.registrationNumber	string	No	CIPC company registration number. max length 50
dealership.vatNumber	string	No	max length 50
dealership.dealerLicenceNumber	string	No	max length 50
dealership.email	string (email)	Yes	
dealership.phone	string	Yes	
dealership.website	string (uri)	No	
dealership.description	string	No	max length 4000
dealership.province	enum	Yes	One of: EASTERN_CAPE, FREE_STATE, GAUTENG, KWAZULU_NATAL, LIMPOPO, MPUMALANGA, NORTHERN_CAPE, NORTH_WEST, WESTERN_CAPE
dealership.city	string	Yes	min length 2; max length 120
dealership.address	string	Yes	min length 3; max length 300
operatingHours	array of objects	No	
operatingHours[].day	enum	Yes	One of: Mon, Tue, Wed, Thu, Fri, Sat, Sun, PublicHoliday
operatingHours[].open	string	No	e.g. "08:00"
operatingHours[].close	string	No	e.g. "17:00"
operatingHours[].closed	boolean	No	default false

GET /api/v1/dealers/{slug}

Public dealer page with branches and opening hours.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 300 s.

Parameter	In	Type	Required	Notes
slug	path	string	Yes	

Content (public) · 9 endpoints

Articles, videos, podcasts, FAQs, static pages and promotions.

GET /api/v1/content/article-categories

Article categories.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 300 s.

GET /api/v1/content/articles

News, reviews, buying advice and guides.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 300 s.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20

Parameter	In	Type	Required	Notes
q	query	string	No	max length 100
category	query	string	No	Category slug. max length 80
type	query	enum	No	One of: NEWS, REVIEW, ADVICE, GUIDE, ARTICLE
tag	query	string	No	max length 60

GET /api/v1/content/articles/{slug}

One published article with its content blocks.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 300 s.

Parameter	In	Type	Required	Notes
slug	path	string	Yes	

GET /api/v1/content/faqs

Frequently asked questions. Filter with ?category=.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 300 s.

Parameter	In	Type	Required	Notes
category	query	string	No	

GET /api/v1/content/media

Videos and podcasts.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 300 s.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
type	query	enum	No	One of: VIDEO, PODCAST

GET /api/v1/content/media/{slug}

One published video or podcast.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 300 s.

Parameter	In	Type	Required	Notes
slug	path	string	Yes	

GET /api/v1/content/pages/{slug}

Static pages: privacy policy, terms, insurance, EV & charging info.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 300 s.

Parameter	In	Type	Required	Notes
slug	path	string	Yes	

GET /api/v1/promotions

Active specials and promotions.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 300 s.

GET /api/v1/promotions/{slug}

One promotion with its vehicles (paginated).

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 300 s.

Parameter	In	Type	Required	Notes
slug	path	string	Yes	
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20

Newsletter · 2 endpoints

Subscribe and unsubscribe.

POST /api/v1/newsletter/subscribe

Subscribe an email address to the newsletter.

Access: Public. Login optional (when signed in, the request is linked to the user). **Success:** 200. Rate limit 5 requests per minute per IP.

Body field	Type	Required	Rules and notes
email	string	Yes	
source	string	No	e.g. "footer"

POST /api/v1/newsletter/unsubscribe

Unsubscribe with the token from any newsletter email.

Access: Public (no login). **Success:** 200.

Body field	Type	Required	Rules and notes
token	string	Yes	

SEO · 4 endpoints

Sitemaps and structured data for the website.

GET /api/v1/seo/sitemap-static.xml

Sitemap of home, makes, models, dealers, articles and pages.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 3600 s.

GET /api/v1/seo/sitemap-vehicles-{page}.xml

Sitemap of vehicle pages, 10 000 per file.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 3600 s.

Parameter	In	Type	Required	Notes
page	path	number	Yes	

GET /api/v1/seo/sitemap.xml

Sitemap index.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 3600 s.

GET /api/v1/seo/vehicles/{slug}/structured-data

schema.org JSON-LD for a vehicle page.

Access: Public (no login). **Success:** 200. Anonymous responses are CDN-cacheable for 3600 s.

Parameter	In	Type	Required	Notes
slug	path	string	Yes	

Customer: account & dashboard · 17 endpoints

Profile, dashboard, favourites, saved searches, history, devices, privacy.

GET /api/v1/me

My profile.

Access: Signed-in user (any role). **Success:** 200.

PATCH /api/v1/me

Update my name, phone or marketing consent.

Access: Signed-in user (any role). **Success:** 200.

Body field	Type	Required	Rules and notes
firstName	string	No	min length 1; max length 80
lastName	string	No	min length 1; max length 80
phone	string	No	
marketingConsent	boolean	No	

GET /api/v1/me/dashboard

Personal dashboard: enquiries, valuations, offers, favourites, saved searches, history, notifications.

Access: Signed-in user (any role). **Success:** 200.

POST /api/v1/me/delete-account

Delete my account: personal data is anonymised.

Access: Signed-in user (any role). **Success:** 204.

Body field	Type	Required	Rules and notes
password	string	Yes	Current password, to confirm. min length 1; max length 128

GET /api/v1/me/export

Download all my personal data.

Access: Signed-in user (any role). **Success:** 200.

GET /api/v1/me/favourites

Saved vehicles, kept across sessions and devices.

Access: Signed-in user (any role). **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20

GET /api/v1/me/favourites/ids

Ids of saved vehicles (to render heart icons).

Access: Signed-in user (any role). **Success:** 200.

PUT /api/v1/me/favourites/{vehicleId}

Save a vehicle to favourites. Safe to call twice.

Access: Signed-in user (any role). **Success:** 200.

Parameter	In	Type	Required	Notes
vehicleId	path	string	Yes	

DELETE /api/v1/me/favourites/{vehicleId}

Remove a vehicle from favourites.

Access: Signed-in user (any role). **Success:** 200.

Parameter	In	Type	Required	Notes
vehicleId	path	string	Yes	

GET /api/v1/me/recently-viewed

Recently viewed vehicles.

Access: Signed-in user (any role). **Success:** 200.

DELETE /api/v1/me/recently-viewed

Clear my recently viewed history.

Access: Signed-in user (any role). **Success:** 200.

GET /api/v1/me/saved-searches

Saved searches.

Access: Signed-in user (any role). **Success:** 200.

POST /api/v1/me/saved-searches

Save the current search filters, with optional alerts.

Access: Signed-in user (any role). **Success:** 201.

Body field	Type	Required	Rules and notes
name	string	Yes	min length 1; max length 120. e.g. "Diesel bakkies under R400k"
criteria	object	Yes	
criteria.q	string	No	Free text over the listing title. max length 100
criteria.make	string	No	Make slugs, comma-separated. max length 500. e.g. "bmw,audi"
criteria.model	string	No	Model slugs; scope to a make with make:model. max length 500. e.g. "bmw:3-series"
criteria.variant	string	No	Variant slugs. max length 500. e.g. "320i-m-sport-2022"
criteria.condition	string	No	NEW, USED, DEMO. max length 50. e.g. "USED"
criteria.category	string	No	Category slugs (FR-25). max length 300. e.g. "suvs,bakkies"
criteria.fuelType	string	No	max length 200. e.g. "PETROL,DIESEL"
criteria.transmission	string	No	max length 50. e.g. "AUTOMATIC"
criteria.drivetrain	string	No	max length 100. e.g. "FOUR_X_FOUR"
criteria.province	string	No	Provinces (FR-27). max length 300. e.g. "GAUTENG,WESTERN_CAPE"
criteria.city	string	No	max length 100
criteria.colour	string	No	max length 200. e.g. "White,Black"

Body field	Type	Required	Rules and notes
<code>criteria.dealer</code>	string	No	Dealer slug. max length 120
<code>criteria.branchId</code>	string (uuid)	No	
<code>criteria.minPrice</code>	number	No	min 0
<code>criteria.maxPrice</code>	number	No	min 0
<code>criteria.minYear</code>	number	No	min 1900
<code>criteria.maxYear</code>	number	No	max 2100
<code>criteria.minMileage</code>	number	No	min 0
<code>criteria.maxMileage</code>	number	No	min 0
<code>criteria.minEngineCc</code>	number	No	min 0
<code>criteria.maxEngineCc</code>	number	No	min 0
<code>criteria.minPowerKw</code>	number	No	min 0
<code>criteria.maxPowerKw</code>	number	No	min 0
<code>criteria.seats</code>	string	No	Seat counts, comma-separated; "8+" for eight or more. max length 50. e.g. "5,7"
<code>criteria.onSpecial</code>	boolean	No	Only vehicles on special (FR-23)
<code>criteria.featured</code>	boolean	No	
<code>criteria.availableOnly</code>	boolean	No	Hide reserved vehicles
<code>criteria.collection</code>	enum	No	One of: <code>hot-sellers</code> , <code>specials</code> , <code>featured</code> , <code>new-arrivals</code> , <code>budget</code> , <code>student</code> , <code>bakkies</code> , <code>electric</code>
<code>criteria.lat</code>	number	No	Centre latitude for radius search
<code>criteria.lng</code>	number	No	Centre longitude for radius search
<code>criteria.radiusKm</code>	number	No	Radius in km (with lat/lng). min 1; max 1000
<code>alertsEnabled</code>	boolean	No	Notify me about new matches and price drops (FR-39). default <code>true</code>

PATCH /api/v1/me/saved-searches/{id}

Rename, change criteria or toggle alerts.

Access: Signed-in user (any role). **Success:** 200.

Parameter	In	Type	Required	Notes
<code>id</code>	path	string	Yes	

Body field	Type	Required	Rules and notes
<code>name</code>	string	No	min length 1; max length 120
<code>criteria</code>	object	No	
<code>criteria.q</code>	string	No	Free text over the listing title. max length 100
<code>criteria.make</code>	string	No	Make slugs, comma-separated. max length 500. e.g. "bmw,audi"
<code>criteria.model</code>	string	No	Model slugs; scope to a make with make:model. max length 500. e.g. "bmw:3-series"
<code>criteria.variant</code>	string	No	Variant slugs. max length 500. e.g. "320i-m-sport-2022"
<code>criteria.condition</code>	string	No	NEW, USED, DEMO. max length 50. e.g. "USED"
<code>criteria.category</code>	string	No	Category slugs (FR-25). max length 300. e.g. "suvs,bakkies"
<code>criteria.fuelType</code>	string	No	max length 200. e.g. "PETROL,DIESEL"

Body field	Type	Required	Rules and notes
criteria.transmission	string	No	max length 50. e.g. "AUTOMATIC"
criteria.drivetrain	string	No	max length 100. e.g. "FOUR_X_FOUR"
criteria.province	string	No	Provinces (FR-27). max length 300. e.g. "GAUTENG, WESTERN_CAPE"
criteria.city	string	No	max length 100
criteria.colour	string	No	max length 200. e.g. "White, Black"
criteria.dealer	string	No	Dealer slug. max length 120
criteria.branchId	string (uuid)	No	
criteria.minPrice	number	No	min 0
criteria.maxPrice	number	No	min 0
criteria.minYear	number	No	min 1900
criteria.maxYear	number	No	max 2100
criteria.minMileage	number	No	min 0
criteria.maxMileage	number	No	min 0
criteria.minEngineCc	number	No	min 0
criteria.maxEngineCc	number	No	min 0
criteria.minPowerKw	number	No	min 0
criteria.maxPowerKw	number	No	min 0
criteria.seats	string	No	Seat counts, comma-separated; "8+" for eight or more. max length 50. e.g. "5, 7"
criteria.onSpecial	boolean	No	Only vehicles on special (FR-23)
criteria.featured	boolean	No	
criteria.availableOnly	boolean	No	Hide reserved vehicles
criteria.collection	enum	No	One of: hot-sellers, specials, featured, new-arrivals, budget, student, bakkies, electric
criteria.lat	number	No	Centre latitude for radius search
criteria.lng	number	No	Centre longitude for radius search
criteria.radiusKm	number	No	Radius in km (with lat/lng). min 1; max 1000
alertsEnabled	boolean	No	

DELETE /api/v1/me/saved-searches/{id}

Delete a saved search.

Access: Signed-in user (any role). **Success:** 204.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

GET /api/v1/me/sessions

Signed-in devices (security settings).

Access: Signed-in user (any role). **Success:** 200.

DELETE /api/v1/me/sessions/{id}

Sign out one device.

Access: Signed-in user (any role). **Success:** 204.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Notifications · 6 endpoints

In-app notifications and notification settings.

GET /api/v1/me/notifications

My in-app notifications.

Access: Signed-in user (any role). **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
unreadOnly	query	boolean	No	

GET /api/v1/me/notifications/preferences

Notification preferences per type and channel.

Access: Signed-in user (any role). **Success:** 200.

PUT /api/v1/me/notifications/preferences

Turn notification channels (in-app, email, SMS) on or off per notification type.

Access: Signed-in user (any role). **Success:** 200.

Body field	Type	Required	Rules and notes
items	array of objects	Yes	max items 200
items[].eventType	string	Yes	e.g. "offer.new"
items[].channel	enum	Yes	One of: IN_APP, EMAIL, SMS
items[].enabled	boolean	Yes	

POST /api/v1/me/notifications/read-all

Mark all my notifications as read.

Access: Signed-in user (any role). **Success:** 201.

GET /api/v1/me/notifications/unread-count

Number of unread notifications. Use it for the bell badge.

Access: Signed-in user (any role). **Success:** 200.

POST /api/v1/me/notifications/{id}/read

Mark one notification as read.

Access: Signed-in user (any role). **Success:** 204.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Customer: enquiries · 4 endpoints

The customer's own enquiries and conversations.

GET `/api/v1/me/enquiries`

My enquiries with status, dealer response and follow-up.

Access: Signed-in user (any role). **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
type	query	enum	No	One of: VEHICLE, QUOTE, BEAT_MY_QUOTE, TRADE_IN, CONCIERGE, HELP_ME_FIND, FINANCE, INSURANCE, TEST_DRIVE, GENERAL
status	query	enum	No	One of: NEW, IN_PROGRESS, RESPONDED, CLOSED, CANCELLED
q	query	string	No	max length 100

GET `/api/v1/me/enquiries/{id}`

One of my enquiries with the full conversation.

Access: Signed-in user (any role). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

POST `/api/v1/me/enquiries/{id}/cancel`

Cancel my enquiry.

Access: Signed-in user (any role). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

POST `/api/v1/me/enquiries/{id}/reply`

Send a follow-up message on my enquiry.

Access: Signed-in user (any role). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
message	string	Yes	min length 1; max length 4000

Customer: my vehicles for sale · 6 endpoints

The customer's valuations and sell requests.

GET `/api/v1/me/sell-requests`

My valuations and sell requests.

Access: Signed-in user (any role). **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20

Parameter	In	Type	Required	Notes
status	query	enum	No	One of: SUBMITTED, UNDER_REVIEW, VALUED, BIDDING, OFFER_ACCEPTED, COMPLETED, CANCELLED, REJECTED
type	query	enum	No	One of: SELL, VALUATION
q	query	string	No	max length 100

GET /api/v1/me/sell-requests/{id}

Valuations, bidding status and every dealer offer with its history.

Access: Signed-in user (any role). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

POST /api/v1/me/sell-requests/{id}/cancel

Cancel my sell request. Open bidding is cancelled too.

Access: Signed-in user (any role). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

POST /api/v1/me/sell-requests/{id}/images

Step 2 of a photo upload: confirm the uploaded photo.

Access: Signed-in user (any role). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
storageKey	string	Yes	min length 10; max length 300
contentType	enum	Yes	One of: image/jpeg, image/png, image/webp

POST /api/v1/me/sell-requests/{id}/images/upload-url

Step 1 of a photo upload: get a presigned upload URL.

Access: Signed-in user (any role). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
contentType	enum	Yes	One of: image/jpeg, image/png, image/webp

POST /api/v1/me/sell-requests/{id}/request-offers

Ask ChangeCars to collect offers from the dealer network.

Access: Signed-in user (any role). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Customer: offers · 4 endpoints

Dealer offers on the customer's vehicles: accept, reject, counter.

GET `/api/v1/me/offers`

Dealer offers on my vehicles.

Access: Signed-in user (any role). **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
status	query	enum	No	One of: SUBMITTED, PENDING, UPDATED, ACCEPTED, REJECTED, EXPIRED, WITHDRAWN, SUPERSEDED

POST `/api/v1/me/offers/{id}/accept`

Accept an offer. Creates the deal and supersedes all other offers atomically.

Access: Signed-in user (any role). **Success:** 200. Supports the Idempotency-Key header.

Parameter	In	Type	Required	Notes
id	path	string	Yes	
Idempotency-Key	header	string	No	Retry-safe acceptance

POST `/api/v1/me/offers/{id}/counter`

Counter an offer with a higher amount.

Access: Signed-in user (any role). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
amount	number	Yes	min 1000; max 100000000
message	string	No	max length 1000

POST `/api/v1/me/offers/{id}/reject`

Reject a dealer's offer.

Access: Signed-in user (any role). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
reason	string	No	max length 1000

8.2 Dealer portal APIs

Used by the dealer app or dealer portal. The signed-in user must be an active member of a dealership. The required dealer permission is shown on each endpoint.

Dealer portal: account · 13 endpoints

GET /api/v1/dealer/branches

List my dealership's branches.

Access: Dealer member. **Success:** 200.

POST /api/v1/dealer/branches

Add a branch.

Access: Dealer member, permission BRANCHES_MANAGE. **Success:** 201.

Body field	Type	Required	Rules and notes
name	string	Yes	min length 2; max length 120. e.g. "Sandton"
email	string (email)	No	
phone	string	No	
address	string	Yes	min length 3; max length 300
city	string	Yes	min length 2; max length 120
province	enum	Yes	One of: EASTERN_CAPE, FREE_STATE, GAUTENG, KWAZULU_NATAL, LIMPOPO, MPUMALANGA, NORTHERN_CAPE, NORTH_WEST, WESTERN_CAPE
latitude	number	No	
longitude	number	No	
operatingHours	array of objects	No	max items 8
operatingHours[].day	enum	Yes	One of: Mon, Tue, Wed, Thu, Fri, Sat, Sun, PublicHoliday
operatingHours[].open	string	No	e.g. "08:00"
operatingHours[].close	string	No	e.g. "17:00"
operatingHours[].closed	boolean	No	default false

PATCH /api/v1/dealer/branches/{id}

Update a branch, or deactivate it with isActive=false.

Access: Dealer member, permission BRANCHES_MANAGE. **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
name	string	No	min length 2; max length 120. e.g. "Sandton"
email	string (email)	No	
phone	string	No	
address	string	No	min length 3; max length 300
city	string	No	min length 2; max length 120
province	enum	No	One of: EASTERN_CAPE, FREE_STATE, GAUTENG, KWAZULU_NATAL, LIMPOPO, MPUMALANGA, NORTHERN_CAPE, NORTH_WEST, WESTERN_CAPE

Body field	Type	Required	Rules and notes
latitude	number	No	
longitude	number	No	
operatingHours	array of objects	No	max items 8
operatingHours[].day	enum	Yes	One of: Mon, Tue, Wed, Thu, Fri, Sat, Sun, PublicHoliday
operatingHours[].open	string	No	e.g. "08:00"
operatingHours[].close	string	No	e.g. "17:00"
operatingHours[].closed	boolean	No	default false
isActive	boolean	No	

GET /api/v1/dealer/documents

List the verification documents my dealership uploaded.

Access: Dealer member, permission DEALER_PROFILE_MANAGE, works even if the dealership is pending, suspended or rejected. **Success:** 200.

POST /api/v1/dealer/documents

Register an uploaded verification document.

Access: Dealer member, permission DEALER_PROFILE_MANAGE, works even if the dealership is pending, suspended or rejected. **Success:** 201.

Body field	Type	Required	Rules and notes
type	enum	Yes	One of: COMPANY_REGISTRATION, TAX_CLEARANCE, OWNER_ID, DEALER_LICENCE, PROOF_OF_ADDRESS, OTHER
fileName	string	Yes	min length 1; max length 200
contentType	enum	Yes	One of: application/pdf, image/jpeg, image/png
storageKey	string	Yes	min length 10; max length 300

POST /api/v1/dealer/documents/upload-url

Presigned URL for a verification document upload.

Access: Dealer member, permission DEALER_PROFILE_MANAGE, works even if the dealership is pending, suspended or rejected. **Success:** 201.

Body field	Type	Required	Rules and notes
type	enum	Yes	One of: COMPANY_REGISTRATION, TAX_CLEARANCE, OWNER_ID, DEALER_LICENCE, PROOF_OF_ADDRESS, OTHER
fileName	string	Yes	min length 1; max length 200
contentType	enum	Yes	One of: application/pdf, image/jpeg, image/png

GET /api/v1/dealer/me

My membership, role and effective permissions.

Access: Dealer member, works even if the dealership is pending, suspended or rejected. **Success:** 200.

GET /api/v1/dealer/profile

My dealership.

Access: Dealer member, works even if the dealership is pending, suspended or rejected. **Success:** 200.

PATCH /api/v1/dealer/profile

Update my dealership details.

Access: Dealer member, permission DEALER_PROFILE_MANAGE. **Success:** 200.

Body field	Type	Required	Rules and notes
name	string	No	min length 2; max length 120. e.g. "Sandton Auto"
legalName	string	No	max length 200
registrationNumber	string	No	CIPC company registration number. max length 50
vatNumber	string	No	max length 50
dealerLicenceNumber	string	No	max length 50
email	string (email)	No	
phone	string	No	
website	string (uri)	No	
description	string	No	max length 4000
province	enum	No	One of: EASTERN_CAPE, FREE_STATE, GAUTENG, KWAZULU_NATAL, LIMPOPO, MPUMALANGA, NORTHERN_CAPE, NORTH_WEST, WESTERN_CAPE
city	string	No	min length 2; max length 120
address	string	No	min length 3; max length 300
logoUrl	string (uri)	No	

GET /api/v1/dealer/staff

List staff members with their roles and effective permissions.

Access: Dealer member, any of STAFF_MANAGE, LEADS_ASSIGN. **Success:** 200.

POST /api/v1/dealer/staff

Create a staff account and email an invitation.

Access: Dealer member, permission STAFF_MANAGE. **Success:** 201.

Body field	Type	Required	Rules and notes
email	string (email)	Yes	
firstName	string	Yes	min length 1; max length 80
lastName	string	Yes	min length 1; max length 80
phone	string	No	
role	enum	Yes	One of: MANAGER, SALES, STAFF
branchId	string (uuid)	No	
extraPermissions	array of enum	No	One of: DEALER_PROFILE_MANAGE, BRANCHES_MANAGE, STAFF_MANAGE, INVENTORY_VIEW, INVENTORY_MANAGE, INVENTORY_PUBLISH, ENQUIRIES_VIEW, ENQUIRIES_RESPOND, LEADS_VIEW_ALL, LEADS_MANAGE_ALL, LEADS_MANAGE_ASSIGNED, LEADS_ASSIGN, BIDDING_VIEW, BIDDING_PARTICIPATE, REPORTS_VIEW

PATCH /api/v1/dealer/staff/{memberId}

Change role, branch, permissions or disable a staff account.

Access: Dealer member, permission STAFF_MANAGE. **Success:** 200.

Parameter	In	Type	Required	Notes
memberId	path	string	Yes	

Body field	Type	Required	Rules and notes
role	enum	No	One of: MANAGER , SALES , STAFF
branchId	string (uuid)	No	
extraPermissions	array of enum	No	One of: DEALER_PROFILE_MANAGE , BRANCHES_MANAGE , STAFF_MANAGE , INVENTORY_VIEW , INVENTORY_MANAGE , INVENTORY_PUBLISH , ENQUIRIES_VIEW , ENQUIRIES_RESPOND , LEADS_VIEW_ALL , LEADS_MANAGE_ALL , LEADS_MANAGE_ASSIGNED , LEADS_ASSIGN , BIDDING_VIEW , BIDDING_PARTICIPATE , REPORTS_VIEW
status	enum	No	One of: ACTIVE , DISABLED

GET /api/v1/dealer/staff/{memberId}/activity

Recent actions performed by one staff member.

Access: Dealer member, permission STAFF_MANAGE. **Success:** 200.

Parameter	In	Type	Required	Notes
memberId	path	string	Yes	
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20

Dealer portal: dashboard · 1 endpoint

GET /api/v1/dealer/dashboard

Dealer dashboard summary.

Access: Dealer member, works even if the dealership is pending, suspended or rejected. **Success:** 200.

Dealer portal: inventory · 20 endpoints

GET /api/v1/dealer/vehicles

My inventory.

Access: Dealer member, permission INVENTORY_VIEW. **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
status	query	array of enum	No	Comma-separated statuses. One of: DRAFT , PENDING_REVIEW , APPROVED , REJECTED , PUBLISHED , RESERVED , SOLD , SUSPENDED , ARCHIVED
branchId	query	string (uuid)	No	
q	query	string	No	max length 100
sort	query	enum	No	One of: recent , price-asc , price-desc , views-desc

POST /api/v1/dealer/vehicles

Create a vehicle in DRAFT.

Access: Dealer member, permission INVENTORY_MANAGE. **Success:** 201.

Body field	Type	Required	Rules and notes
makeId	string (uuid)	Yes	
modelId	string (uuid)	Yes	
generationId	string (uuid)	No	
variantId	string (uuid)	No	Strongly recommended: fills specs from the catalogue
branchId	string (uuid)	No	
condition	enum	Yes	One of: NEW , USED , DEMO
year	number	Yes	min 1900; max 2027. e.g. 2021
mileage	number	Yes	min 0; max 2000000. e.g. 45000
price	number	Yes	Rand. min 1000; max 100000000. e.g. 459900
title	string	No	Defaults to "<year> <make> <model> <variant>". min length 3; max length 160
description	string	No	max length 10000
stockNumber	string	No	max length 60
transmission	enum	No	One of: MANUAL , AUTOMATIC
fuelType	enum	No	One of: PETROL , DIESEL , HYBRID , PLUGIN_HYBRID , ELECTRIC , LPG , OTHER
drivetrain	enum	No	One of: FWD , RWD , AWD , FOUR_X_TWO , FOUR_X_FOUR
colour	string	No	max length 40
engineCapacityCc	number	No	min 0; max 20000
powerKw	number	No	min 0; max 2000
cylinders	number	No	min 0; max 32
seats	number	No	min 1; max 60
doors	number	No	min 0; max 6
vin	string	No	Private, never shown publicly
registrationNumber	string	No	Private, never shown publicly. max length 20
province	enum	No	Defaults to the branch/dealer province. One of: EASTERN_CAPE , FREE_STATE , GAUTENG , KWAZULU_NATAL , LIMPOPO , MPUMALANGA , NORTHERN_CAPE , NORTH_WEST , WESTERN_CAPE
city	string	No	max length 120
latitude	number	No	
longitude	number	No	
categoryIds	array of string	No	Extra category ids; body and fuel categories are assigned automatically. max items 10
featureIds	array of string	No	Vehicle-level feature ids (FR-50). max items 200
isSpecial	boolean	No	
specialPrice	number	No	Promotional price when on special (FR-23). min 1000
promotionId	string (uuid)	No	

GET `/api/v1/dealer/vehicles/{id}`

One vehicle from my inventory, with photos, history and allowed next actions.

Access: Dealer member, permission INVENTORY_VIEW. **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

PATCH /api/v1/dealer/vehicles/{id}

Edit a vehicle. After approval only price, specials, description, colour, branch, stock number and features may change.

Access: Dealer member, permission INVENTORY_MANAGE. **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
makeId	string (uuid)	No	
modelId	string (uuid)	No	
generationId	string (uuid)	No	
variantId	string (uuid)	No	Strongly recommended: fills specs from the catalogue
branchId	string (uuid)	No	
condition	enum	No	One of: NEW , USED , DEMO
year	number	No	min 1900; max 2027. e.g. 2021
mileage	number	No	min 0; max 2000000. e.g. 45000
price	number	No	Rand. min 1000; max 100000000. e.g. 459900
title	string	No	Defaults to "<year> <make> <model> <variant>". min length 3; max length 160
description	string	No	max length 10000
stockNumber	string	No	max length 60
transmission	enum	No	One of: MANUAL , AUTOMATIC
fuelType	enum	No	One of: PETROL , DIESEL , HYBRID , PLUGIN_HYBRID , ELECTRIC , LPG , OTHER
drivetrain	enum	No	One of: FWD , RWD , AWD , FOUR_X_TWO , FOUR_X_FOUR
colour	string	No	max length 40
engineCapacityCc	number	No	min 0; max 20000
powerKw	number	No	min 0; max 2000
cylinders	number	No	min 0; max 32
seats	number	No	min 1; max 60
doors	number	No	min 0; max 6
vin	string	No	Private, never shown publicly
registrationNumber	string	No	Private, never shown publicly. max length 20
province	enum	No	Defaults to the branch/dealer province. One of: EASTERN_CAPE , FREE_STATE , GAUTENG , KWAZULU_NATAL , LIMPOPO , MPUMALANGA , NORTHERN_CAPE , NORTH_WEST , WESTERN_CAPE
city	string	No	max length 120
latitude	number	No	
longitude	number	No	
categoryIds	array of string	No	Extra category ids; body and fuel categories are assigned automatically. max items 10
featureIds	array of string	No	Vehicle-level feature ids (FR-50). max items 200
isSpecial	boolean	No	
specialPrice	number	No	Promotional price when on special (FR-23). min 1000
promotionId	string (uuid)	No	

Body field	Type	Required	Rules and notes
<code>expectedVersion</code>	number	No	Optimistic concurrency: the version you last read

POST /api/v1/dealer/vehicles/{id}/archive

Remove a listing (kept for history).

Access: Dealer member, permission INVENTORY_MANAGE. **Success:** 201.

Parameter	In	Type	Required	Notes
<code>id</code>	path	string	Yes	

Body field	Type	Required	Rules and notes
<code>reason</code>	string	No	Reason (required for reject and admin suspend). max length 1000

POST /api/v1/dealer/vehicles/{id}/edit

Move back to DRAFT for material edits (requires a new review).

Access: Dealer member, permission INVENTORY_MANAGE. **Success:** 201.

Parameter	In	Type	Required	Notes
<code>id</code>	path	string	Yes	

GET /api/v1/dealer/vehicles/{id}/images

List the photos of a vehicle.

Access: Dealer member, permission INVENTORY_VIEW. **Success:** 200.

Parameter	In	Type	Required	Notes
<code>id</code>	path	string	Yes	

PUT /api/v1/dealer/vehicles/{id}/images/order

Set photo order. The first photo becomes the main photo.

Access: Dealer member, permission INVENTORY_MANAGE. **Success:** 200.

Parameter	In	Type	Required	Notes
<code>id</code>	path	string	Yes	

Body field	Type	Required	Rules and notes
<code>imageIds</code>	array of string	Yes	Image ids in display order; the first becomes primary. max items 100

POST /api/v1/dealer/vehicles/{id}/images/upload-url

Step 1: get a presigned URL, then PUT the file to it.

Access: Dealer member, permission INVENTORY_MANAGE. **Success:** 201.

Parameter	In	Type	Required	Notes
<code>id</code>	path	string	Yes	

Body field	Type	Required	Rules and notes
<code>contentType</code>	enum	Yes	One of: <code>image/jpeg</code> , <code>image/png</code> , <code>image/webp</code>
<code>sizeBytes</code>	number	Yes	File size in bytes. min 1

PATCH /api/v1/dealer/vehicles/{id}/images/{imageId}

Set alt text or make a photo the main photo.

Access: Dealer member, permission INVENTORY_MANAGE. **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	
imageId	path	string	Yes	

Body field	Type	Required	Rules and notes
altText	string	No	max length 200
isPrimary	boolean	No	

DELETE /api/v1/dealer/vehicles/{id}/images/{imageId}

Delete a photo.

Access: Dealer member, permission INVENTORY_MANAGE. **Success:** 204.

Parameter	In	Type	Required	Notes
id	path	string	Yes	
imageId	path	string	Yes	

POST /api/v1/dealer/vehicles/{id}/images/{imageId}/complete

Step 2: confirm the upload; resizing happens in the background.

Access: Dealer member, permission INVENTORY_MANAGE. **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	
imageId	path	string	Yes	

POST /api/v1/dealer/vehicles/{id}/publish

Publish an approved vehicle.

Access: Dealer member, permission INVENTORY_PUBLISH, dealership must be APPROVED. **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

POST /api/v1/dealer/vehicles/{id}/release

Cancel a reservation. The vehicle becomes available again.

Access: Dealer member, permission INVENTORY_MANAGE. **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

POST /api/v1/dealer/vehicles/{id}/reserve

Reserve a published vehicle (stays visible as reserved, BR-05).

Access: Dealer member, permission INVENTORY_MANAGE. **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
reason	string	No	Reason (required for reject and admin suspend). max length 1000
reservedUntil	string	No	Reservation expiry; defaults to RESERVATION_HOLD_HOURS from now

POST /api/v1/dealer/vehicles/{id}/sell

Mark as sold (removed from search results, BR-04).

Access: Dealer member, permission INVENTORY_MANAGE. **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
reason	string	No	Reason (required for reject and admin suspend). max length 1000

POST /api/v1/dealer/vehicles/{id}/submit

Submit for admin review (DRAFT/REJECTED → PENDING_REVIEW).

Access: Dealer member, permission INVENTORY_MANAGE, dealership must be APPROVED. **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

POST /api/v1/dealer/vehicles/{id}/suspend

Temporarily take a listing offline.

Access: Dealer member, permission INVENTORY_MANAGE. **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
reason	string	No	Reason (required for reject and admin suspend). max length 1000

POST /api/v1/dealer/vehicles/{id}/unsuspend

Put a suspended listing back online.

Access: Dealer member, permission INVENTORY_MANAGE, dealership must be APPROVED. **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

POST /api/v1/dealer/vehicles/{id}/withdraw

Withdraw from review (PENDING_REVIEW → DRAFT).

Access: Dealer member, permission INVENTORY_MANAGE. **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Dealer portal: enquiries · 3 endpoints

GET /api/v1/dealer/enquiries

Enquiries sent to my dealership (or my branch).

Access: Dealer member, permission ENQUIRIES_VIEW. **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
type	query	enum	No	One of: VEHICLE, QUOTE, BEAT_MY_QUOTE, TRADE_IN, CONCIERGE, HELP_ME_FIND, FINANCE, INSURANCE, TEST_DRIVE, GENERAL
status	query	enum	No	One of: NEW, IN_PROGRESS, RESPONDED, CLOSED, CANCELLED
q	query	string	No	max length 100

GET /api/v1/dealer/enquiries/{id}

One enquiry with the conversation and its lead.

Access: Dealer member, permission ENQUIRIES_VIEW. **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

POST /api/v1/dealer/enquiries/{id}/respond

Reply to the customer; also moves the lead from NEW to CONTACTED.

Access: Dealer member, permission ENQUIRIES_RESPOND. **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
message	string	Yes	min length 1; max length 4000
status	enum	No	One of: IN_PROGRESS, RESPONDED, CLOSED; default "RESPONDED"

Dealer portal: CRM · 8 endpoints

GET /api/v1/dealer/leads

Leads visible to me: all (LEADS_VIEW_ALL) or only assigned to me.

Access: Dealer member, any of LEADS_VIEW_ALL, LEADS_MANAGE_ALL, LEADS_MANAGE_ASSIGNED. **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
stage	query	enum	No	One of: NEW, CONTACTED, QUALIFIED, NEGOTIATION, OFFER_SENT, WON, LOST
assignedTo	query	string	No	"me", "unassigned" or a member id. max length 40
branchId	query	string (uuid)	No	
source	query	enum	No	One of: WEBSITE_ENQUIRY, QUOTE_REQUEST, TRADE_IN, OFFER_ACCEPTED, PHONE, WHATSAPP, EMAIL, WALK_IN, OTHER

Parameter	In	Type	Required	Notes
q	query	string	No	max length 100
followUpDue	query	boolean	No	Only leads with a follow-up due now or earlier

POST /api/v1/dealer/leads

Create a lead manually (walk-in, phone, WhatsApp...).

Access: Dealer member, any of LEADS_MANAGE_ALL, LEADS_MANAGE_ASSIGNED. **Success:** 201.

Body field	Type	Required	Rules and notes
name	string	Yes	min length 1; max length 160
email	string (email)	No	
phone	string	No	
source	enum	Yes	One of: PHONE, WHATSAPP, EMAIL, WALK_IN, OTHER
vehicleId	string (uuid)	No	
branchId	string (uuid)	No	
assignedToId	string (uuid)	No	
note	string	No	max length 4000
nextFollowUpAt	string	No	

GET /api/v1/dealer/leads/stats

Pipeline counts, conversion rate, average first response time, overdue follow-ups.

Access: Dealer member, any of LEADS_VIEW_ALL, LEADS_MANAGE_ALL, LEADS_MANAGE_ASSIGNED. **Success:** 200.

GET /api/v1/dealer/leads/{id}

One lead with its enquiry, vehicle, deal and full activity history.

Access: Dealer member, any of LEADS_VIEW_ALL, LEADS_MANAGE_ALL, LEADS_MANAGE_ASSIGNED. **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

PATCH /api/v1/dealer/leads/{id}

Update lead contact details, vehicle or next follow-up date.

Access: Dealer member, any of LEADS_MANAGE_ALL, LEADS_MANAGE_ASSIGNED. **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
name	string	No	min length 1; max length 160
email	string (email)	No	
phone	string	No	
vehicleId	string (uuid)	No	
nextFollowUpAt	string	No	

POST /api/v1/dealer/leads/{id}/activities

Log a call, email, WhatsApp, meeting or note and schedule the next follow-up.

Access: Dealer member, any of LEADS_MANAGE_ALL, LEADS_MANAGE_ASSIGNED. **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
type	enum	Yes	One of: NOTE , CALL , EMAIL , SMS , WHATSAPP , MEETING
note	string	Yes	min length 1; max length 4000
nextFollowUpAt	string	No	Schedule the next follow-up

POST /api/v1/dealer/leads/{id}/assign

Assign a lead to a staff member, or unassign it with null.

Access: Dealer member, permission LEADS_ASSIGN. **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
memberId	string (uuid)	Yes	Dealer member id, or null to unassign

POST /api/v1/dealer/leads/{id}/stage

Move through New → Contacted → Qualified → Negotiation → Offer Sent → Won/Lost.

Access: Dealer member, any of LEADS_MANAGE_ALL, LEADS_MANAGE_ASSIGNED. **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
stage	enum	Yes	One of: NEW , CONTACTED , QUALIFIED , NEGOTIATION , OFFER_SENT , WON , LOST
note	string	No	max length 2000
lostReason	string	No	Required when stage is LOST. max length 500

Dealer portal: bidding · 7 endpoints

GET /api/v1/dealer/bidding/sessions

Vehicles open for bidding that my dealership is eligible for.

Access: Dealer member, permission BIDDING_VIEW, dealership must be APPROVED. **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
status	query	enum	No	One of: SCHEDULED , OPEN , CLOSED , AWARDED , CANCELLED

GET /api/v1/dealer/bidding/sessions/{id}

One bidding session with vehicle details and my offer history.

Access: Dealer member, permission BIDDING_VIEW, dealership must be APPROVED. **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

POST /api/v1/dealer/bidding/sessions/{id}/offers

Submit (or revise/renew) this dealership's offer.

Access: Dealer member, permission BIDDING_PARTICIPATE, dealership must be APPROVED. **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
amount	number	Yes	Offer in rand. min 1000; max 100000000
terms	string	No	Additional terms shown to the customer. max length 2000
dealerNotes	string	No	Internal dealer notes (never shown to the customer). max length 2000

GET /api/v1/dealer/offers

My dealership's offers.

Access: Dealer member, permission BIDDING_VIEW. **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
status	query	enum	No	One of: SUBMITTED, PENDING, UPDATED, ACCEPTED, REJECTED, EXPIRED, WITHDRAWN, SUPERSEDED

PATCH /api/v1/dealer/offers/{id}

Revise an offer; full history is kept.

Access: Dealer member, permission BIDDING_PARTICIPATE, dealership must be APPROVED. **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
amount	number	Yes	Offer in rand. min 1000; max 100000000
terms	string	No	Additional terms shown to the customer. max length 2000
dealerNotes	string	No	Internal dealer notes (never shown to the customer). max length 2000
expectedVersion	number	No	Optimistic concurrency: the offer version you last saw

POST /api/v1/dealer/offers/{id}/counter-response

Accept or decline the customer's counter-offer.

Access: Dealer member, permission BIDDING_PARTICIPATE. **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
accept	boolean	Yes	true = accept the customer counter amount
message	string	No	max length 1000

POST /api/v1/dealer/offers/{id}/withdraw

Withdraw my offer (only if the session allows withdrawals).

Access: Dealer member, permission BIDDING_PARTICIPATE. **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

8.3 Admin panel APIs

Used by the ChangeCars admin panel. Requires the ADMIN or SUPER_ADMIN role.

Admin: dealers · 4 endpoints

GET /api/v1/admin/dealers

List dealers. Filter by status, province or text.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
q	query	string	No	max length 100
province	query	enum	No	One of: EASTERN_CAPE, FREE_STATE, GAUTENG, KWAZULU_NATAL, LIMPOPO, MPUMALANGA, NORTHERN_CAPE, NORTH_WEST, WESTERN_CAPE
status	query	enum	No	One of: PENDING, APPROVED, REJECTED, SUSPENDED

GET /api/v1/admin/dealers/{id}

Full dealer record: branches, members, documents (with download links) and stats.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

PATCH /api/v1/admin/dealers/{id}

Update plan, bidding eligibility or rating.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
plan	enum	No	One of: BASIC, SILVER, GOLD
biddingEnabled	boolean	No	
rating	number	No	min 0; max 5

PATCH /api/v1/admin/dealers/{id}/status

Approve, reject, suspend or reinstate a dealer.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
status	enum	Yes	One of: APPROVED , REJECTED , SUSPENDED
reason	string	No	Required when rejecting or suspending. max length 1000

Admin: vehicles · 12 endpoints

GET /api/v1/admin/vehicles

List all vehicles. Filter by dealer, status or text.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
status	query	array of enum	No	Comma-separated statuses. One of: DRAFT , PENDING_REVIEW , APPROVED , REJECTED , PUBLISHED , RESERVED , SOLD , SUSPENDED , ARCHIVED
branchId	query	string (uuid)	No	
q	query	string	No	max length 100
sort	query	enum	No	One of: recent , price-asc , price-desc , views-desc
dealerId	query	string (uuid)	No	

GET /api/v1/admin/vehicles/review-queue

Listings awaiting approval, oldest first.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20

GET /api/v1/admin/vehicles/{id}

Full vehicle record including private fields and history.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

PATCH /api/v1/admin/vehicles/{id}

Edit a vehicle as an administrator (same edit rules as dealers).

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
makeId	string (uuid)	No	

Body field	Type	Required	Rules and notes
modelId	string (uuid)	No	
generationId	string (uuid)	No	
variantId	string (uuid)	No	Strongly recommended: fills specs from the catalogue
branchId	string (uuid)	No	
condition	enum	No	One of: NEW , USED , DEMO
year	number	No	min 1900; max 2027. e.g. 2021
mileage	number	No	min 0; max 2000000. e.g. 45000
price	number	No	Rand. min 1000; max 100000000. e.g. 459900
title	string	No	Defaults to "<year> <make> <model> <variant>". min length 3; max length 160
description	string	No	max length 10000
stockNumber	string	No	max length 60
transmission	enum	No	One of: MANUAL , AUTOMATIC
fuelType	enum	No	One of: PETROL , DIESEL , HYBRID , PLUGIN_HYBRID , ELECTRIC , LPG , OTHER
drivetrain	enum	No	One of: FWD , RWD , AWD , FOUR_X_TWO , FOUR_X_FOUR
colour	string	No	max length 40
engineCapacityCc	number	No	min 0; max 20000
powerKw	number	No	min 0; max 2000
cylinders	number	No	min 0; max 32
seats	number	No	min 1; max 60
doors	number	No	min 0; max 6
vin	string	No	Private, never shown publicly
registrationNumber	string	No	Private, never shown publicly. max length 20
province	enum	No	Defaults to the branch/dealer province. One of: EASTERN_CAPE , FREE_STATE , GAUTENG , KWAZULU_NATAL , LIMPOPO , MPUMALANGA , NORTHERN_CAPE , NORTH_WEST , WESTERN_CAPE
city	string	No	max length 120
latitude	number	No	
longitude	number	No	
categoryIds	array of string	No	Extra category ids; body and fuel categories are assigned automatically. max items 10
featureIds	array of string	No	Vehicle-level feature ids (FR-50). max items 200
isSpecial	boolean	No	
specialPrice	number	No	Promotional price when on special (FR-23). min 1000
promotionId	string (uuid)	No	
expectedVersion	number	No	Optimistic concurrency: the version you last read

POST /api/v1/admin/vehicles/{id}/approve

Approve a listing that is waiting for review.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
reason	string	No	Reason (required for reject and admin suspend). max length 1000

POST /api/v1/admin/vehicles/{id}/archive

Archive a listing.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
reason	string	No	Reason (required for reject and admin suspend). max length 1000

PATCH /api/v1/admin/vehicles/{id}/flags

Feature a vehicle or mark it as special.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
isFeatured	boolean	No	
isSpecial	boolean	No	

POST /api/v1/admin/vehicles/{id}/publish

Publish an approved listing on behalf of the dealer.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

POST /api/v1/admin/vehicles/{id}/reject

Reject a listing. A reason is required.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
reason	string	No	Reason (required for reject and admin suspend). max length 1000

POST /api/v1/admin/vehicles/{id}/relist

Correct a mistaken sale: SOLD → PUBLISHED.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
reason	string	No	Reason (required for reject and admin suspend). max length 1000

POST /api/v1/admin/vehicles/{id}/suspend

Take a listing offline. A reason is required. Only an admin can bring it back.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
reason	string	No	Reason (required for reject and admin suspend). max length 1000

POST /api/v1/admin/vehicles/{id}/unsuspend

Put a suspended listing back online.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Admin: catalogue · 24 endpoints

GET /api/v1/admin/catalogue/categories

List all vehicle categories.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

POST /api/v1/admin/catalogue/categories

Create a vehicle category.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Body field	Type	Required	Rules and notes
name	string	Yes	min length 1; max length 80. e.g. "SUVs"
description	string	No	max length 500
icon	string	No	max length 80
sortOrder	number	No	
isActive	boolean	No	

PATCH /api/v1/admin/catalogue/categories/{id}

Update a vehicle category.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
name	string	No	min length 1; max length 80. e.g. "SUVs"
description	string	No	max length 500
icon	string	No	max length 80
sortOrder	number	No	
isActive	boolean	No	

DELETE /api/v1/admin/catalogue/categories/{id}

Delete a vehicle category.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 204.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

POST /api/v1/admin/catalogue/feature-assignments

Assign a standard/optional feature to a make, model, generation or variant.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Body field	Type	Required	Rules and notes
featureId	string (uuid)	Yes	
availability	enum	Yes	One of: STANDARD, OPTIONAL
level	enum	Yes	One of: make, model, generation, variant
targetId	string (uuid)	Yes	

DELETE /api/v1/admin/catalogue/feature-assignments/{id}

Remove a feature from a make, model, generation or variant.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 204.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

POST /api/v1/admin/catalogue/features

Create a feature.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Body field	Type	Required	Rules and notes
name	string	Yes	min length 1; max length 120. e.g. "Adaptive cruise control"
group	enum	No	One of: SAFETY, COMFORT, TECHNOLOGY, EXTERIOR, INTERIOR, PERFORMANCE, OTHER
description	string	No	max length 500

PATCH /api/v1/admin/catalogue/features/{id}

Update a feature.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
name	string	No	min length 1; max length 120. e.g. "Adaptive cruise control"
group	enum	No	One of: SAFETY, COMFORT, TECHNOLOGY, EXTERIOR, INTERIOR, PERFORMANCE, OTHER
description	string	No	max length 500

DELETE /api/v1/admin/catalogue/features/{id}

Delete a feature.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 204.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

POST /api/v1/admin/catalogue/generations

Create a generation (for example BMW 3 Series G20).

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Body field	Type	Required	Rules and notes
modelId	string (uuid)	Yes	
name	string	Yes	min length 1; max length 80. e.g. "G20"
yearFrom	number	Yes	min 1900; max 2028. e.g. 2019
yearTo	number	No	min 1900; max 2028. e.g. 2025

PATCH /api/v1/admin/catalogue/generations/{id}

Update a generation.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
name	string	No	min length 1; max length 80. e.g. "G20"
yearFrom	number	No	min 1900; max 2028. e.g. 2019
yearTo	number	No	min 1900; max 2028. e.g. 2025

DELETE /api/v1/admin/catalogue/generations/{id}

Delete a generation.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 204.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

GET /api/v1/admin/catalogue/makes

List all makes, including inactive ones.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

POST /api/v1/admin/catalogue/makes

Create a make.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Body field	Type	Required	Rules and notes
name	string	Yes	min length 1; max length 80. e.g. "BMW"
logoUrl	string (uri)	No	
country	string	No	max length 80. e.g. "Germany"
sortOrder	number	No	
isActive	boolean	No	

PATCH /api/v1/admin/catalogue/makes/{id}

Update a make.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
name	string	No	min length 1; max length 80. e.g. "BMW"
logoUrl	string (uri)	No	
country	string	No	max length 80. e.g. "Germany"
sortOrder	number	No	
isActive	boolean	No	

DELETE /api/v1/admin/catalogue/makes/{id}

Delete a make (fails with 409 if models or vehicles reference it; deactivate instead).

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 204.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

GET /api/v1/admin/catalogue/makes/{makeId}/models

List the models of a make.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
makeId	path	string	Yes	

POST /api/v1/admin/catalogue/models

Create a model.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Body field	Type	Required	Rules and notes
makeId	string (uuid)	Yes	
name	string	Yes	min length 1; max length 80. e.g. "3 Series"
defaultCategoryId	string (uuid)	No	
isActive	boolean	No	

PATCH /api/v1/admin/catalogue/models/{id}

Update a model.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
name	string	No	min length 1; max length 80. e.g. "3 Series"
defaultCategoryId	string (uuid)	No	
isActive	boolean	No	

DELETE /api/v1/admin/catalogue/models/{id}

Delete a model. Fails with 409 if variants or vehicles still use it.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 204.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

GET /api/v1/admin/catalogue/models/{modelId}/variants

List the variants of a model with their specifications.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
modelId	path	string	Yes	

POST /api/v1/admin/catalogue/variants

Create a variant, optionally with its full specification.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Body field	Type	Required	Rules and notes
modelId	string (uuid)	Yes	
generationId	string (uuid)	No	
name	string	Yes	min length 1; max length 120. e.g. "320i M Sport"
yearFrom	number	Yes	min 1900; max 2028. e.g. 2022
yearTo	number	No	min 1900; max 2028
bodyCategoryId	string (uuid)	No	
basePrice	number	No	List price in rand (new vehicles). min 0
isActive	boolean	No	
specification	object	No	
specification.engine	string	No	max length 120. e.g. "2.0L 4-cylinder turbo petrol"
specification.engineCapacity	number	No	min 0; max 20000
specification.cylinders	number	No	min 0; max 32
specification.powerKw	number	No	min 0; max 2000
specification.torqueNm	number	No	min 0; max 3000

Body field	Type	Required	Rules and notes
specification.transmission	enum	No	One of: MANUAL, AUTOMATIC
specification.gears	number	No	min 0; max 12
specification.drivetrain	enum	No	One of: FWD, RWD, AWD, FOUR_X_TWO, FOUR_X_FOUR
specification.fuelType	enum	No	One of: PETROL, DIESEL, HYBRID, PLUGIN_HYBRID, ELECTRIC, LPG, OTHER
specification.fuelConsumptionL100	number	No	min 0; max 50
specification.co2GKm	number	No	min 0; max 1000
specification.seats	number	No	min 1; max 60
specification.doors	number	No	min 0; max 6
specification.lengthMm	number	No	min 0; max 30000
specification.widthMm	number	No	min 0; max 5000
specification.heightMm	number	No	min 0; max 5000
specification.wheelbaseMm	number	No	min 0; max 10000
specification.bootLitres	number	No	min 0; max 10000
specification.kerbWeightKg	number	No	min 0; max 50000
specification.fuelTankLitres	number	No	min 0; max 1000
specification.topSpeedKmh	number	No	min 0; max 500
specification.zeroTo100Sec	number	No	min 0; max 60
specification.batteryKwh	number	No	min 0; max 500
specification.electricRangeKm	number	No	min 0; max 2000
specification.warranty	string	No	max length 200. e.g. "5 years / 100 000 km"
specification.servicePlan	string	No	max length 200. e.g. "5 years / 100 000 km"
specification.safetyRating	string	No	max length 100. e.g. "5-star Euro NCAP"
specification.extra	object	No	Other technical specifications as key/value pairs

PATCH /api/v1/admin/catalogue/variants/{id}

Update a variant and/or its specification.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
generationId	string (uuid)	No	
name	string	No	min length 1; max length 120. e.g. "320i M Sport"
yearFrom	number	No	min 1900; max 2028. e.g. 2022
yearTo	number	No	min 1900; max 2028
bodyCategoryId	string (uuid)	No	
basePrice	number	No	List price in rand (new vehicles). min 0
isActive	boolean	No	
specification	object	No	

Body field	Type	Required	Rules and notes
specification.engine	string	No	max length 120. e.g. "2.0L 4-cylinder turbo petrol"
specification.engineCapacityCc	number	No	min 0; max 20000
specification.cylinders	number	No	min 0; max 32
specification.powerKw	number	No	min 0; max 2000
specification.torqueNm	number	No	min 0; max 3000
specification.transmission	enum	No	One of: MANUAL, AUTOMATIC
specification.gears	number	No	min 0; max 12
specification.drivetrain	enum	No	One of: FWD, RWD, AWD, FOUR_X_TWO, FOUR_X_FOUR
specification.fuelType	enum	No	One of: PETROL, DIESEL, HYBRID, PLUGIN_HYBRID, ELECTRIC, LPG, OTHER
specification.fuelConsumptionL100	number	No	min 0; max 50
specification.co2GKm	number	No	min 0; max 1000
specification.seats	number	No	min 1; max 60
specification.doors	number	No	min 0; max 6
specification.lengthMm	number	No	min 0; max 30000
specification.widthMm	number	No	min 0; max 5000
specification.heightMm	number	No	min 0; max 5000
specification.wheelbaseMm	number	No	min 0; max 10000
specification.bootLitres	number	No	min 0; max 10000
specification.kerbWeightKg	number	No	min 0; max 50000
specification.fuelTankLitres	number	No	min 0; max 1000
specification.topSpeedKmh	number	No	min 0; max 500
specification.zeroTo100Sec	number	No	min 0; max 60
specification.batteryKwh	number	No	min 0; max 500
specification.electricRangeKm	number	No	min 0; max 2000
specification.warranty	string	No	max length 200. e.g. "5 years / 100 000 km"
specification.servicePlan	string	No	max length 200. e.g. "5 years / 100 000 km"
specification.safetyRating	string	No	max length 100. e.g. "5-star Euro NCAP"
specification.extra	object	No	Other technical specifications as key/value pairs

DELETE /api/v1/admin/catalogue/variants/{id}

Delete a variant.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 204.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Admin: enquiries · 4 endpoints

GET /api/v1/admin/enquiries

All enquiries. Use platformOnly=true for requests handled by the ChangeCars team.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
type	query	enum	No	One of: VEHICLE, QUOTE, BEAT_MY_QUOTE, TRADE_IN, CONCIERGE, HELP_ME_FIND, FINANCE, INSURANCE, TEST_DRIVE, GENERAL
status	query	enum	No	One of: NEW, IN_PROGRESS, RESPONDED, CLOSED, CANCELLED
q	query	string	No	max length 100
dealerId	query	string (uuid)	No	
platformOnly	query	boolean	No	Only enquiries handled by the platform team (no dealer)

GET /api/v1/admin/enquiries/{id}

One enquiry with customer, dealer, vehicle and conversation.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

PATCH /api/v1/admin/enquiries/{id}

Change status, assign a team member or route the request to a dealer.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
status	enum	No	One of: NEW, IN_PROGRESS, RESPONDED, CLOSED, CANCELLED
assignedAdminId	string (uuid)	No	Platform team member handling it
dealerId	string (uuid)	No	Route to a dealer (creates a lead for them)

POST /api/v1/admin/enquiries/{id}/respond

Reply to an enquiry as the ChangeCars team.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
message	string	Yes	min length 1; max length 4000
status	enum	No	One of: IN_PROGRESS, RESPONDED, CLOSED; default "RESPONDED"

Admin: sell requests · 4 endpoints

GET /api/v1/admin/sell-requests

List sell and valuation requests.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
status	query	enum	No	One of: SUBMITTED, UNDER_REVIEW, VALUED, BIDDING, OFFER_ACCEPTED, COMPLETED, CANCELLED, REJECTED
type	query	enum	No	One of: SELL, VALUATION
q	query	string	No	max length 100

GET /api/v1/admin/sell-requests/{id}

Full sell request with valuations, bidding sessions, offers and deal.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

PATCH /api/v1/admin/sell-requests/{id}/status

Change the status (review, reject, complete, cancel).

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
status	enum	Yes	One of: UNDER_REVIEW, REJECTED, COMPLETED, CANCELLED
reason	string	No	max length 1000

POST /api/v1/admin/sell-requests/{id}/valuations

Record a manual valuation.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
estimateLow	number	Yes	min 0
estimateMid	number	Yes	min 0
estimateHigh	number	Yes	min 0
notes	string	No	max length 2000

Admin: bidding · 4 endpoints

GET `/api/v1/admin/bidding-sessions`

List bidding sessions.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
status	query	enum	No	One of: SCHEDULED, OPEN, CLOSED, AWARDED, CANCELLED

POST `/api/v1/admin/bidding-sessions/{id}/cancel`

Cancel a bidding process. Active offers are rejected.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
reason	string	No	max length 1000

POST `/api/v1/admin/bidding-sessions/{id}/reopen`

Explicitly reopen a closed bidding process.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
closesAt	string	Yes	
reason	string	No	min length 1; max length 500

POST `/api/v1/admin/sell-requests/{id}/bidding-sessions`

Publish a vehicle to the dealer network with explicit bidding rules.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
opensAt	string	No	Defaults to now. A future time schedules the session.
closesAt	string	Yes	When bidding closes (BR-07)
allowBidModification	boolean	No	Dealers may revise their bid (BR-08). default true
allowBidWithdrawal	boolean	No	default true
allowCounterOffers	boolean	No	Customer may counter an offer (FR-56). default true
offerValidityHours	number	No	Offer validity in hours (BR-10). min 1; max 720; default 48
reservePrice	number	No	Minimum acceptable bid (rand). min 0

Body field	Type	Required	Rules and notes
<code>eligibleProvinces</code>	array of enum	No	Only dealers in these provinces (BR-06). Empty = all.. One of: <code>EASTERN_CAPE</code> , <code>FREE_STATE</code> , <code>GAUTENG</code> , <code>KWAZULU_NATAL</code> , <code>LIMPOPO</code> , <code>MPUMALANGA</code> , <code>NORTHERN_CAPE</code> , <code>NORTH_WEST</code> , <code>WESTERN_CAPE</code>
<code>inviteDealerIds</code>	array of string	No	Invite-only: only these dealers may view and bid. max items 500

Admin: content · 28 endpoints

POST `/api/v1/admin/article-categories`

Create an article category.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Body field	Type	Required	Rules and notes
<code>name</code>	string	Yes	min length 2; max length 80

DELETE `/api/v1/admin/article-categories/{id}`

Delete an article category.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 204.

Parameter	In	Type	Required	Notes
<code>id</code>	path	string	Yes	

GET `/api/v1/admin/articles`

List articles (all statuses).

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
<code>page</code>	query	number	No	min 1; default 1
<code>pageSize</code>	query	number	No	min 1; max 100; default 20
<code>q</code>	query	string	No	max length 100
<code>category</code>	query	string	No	Category slug. max length 80
<code>type</code>	query	enum	No	One of: <code>NEWS</code> , <code>REVIEW</code> , <code>ADVICE</code> , <code>GUIDE</code> , <code>ARTICLE</code>
<code>tag</code>	query	string	No	max length 60
<code>status</code>	query	enum	No	One of: <code>DRAFT</code> , <code>PUBLISHED</code> , <code>ARCHIVED</code>

POST `/api/v1/admin/articles`

Create an article (starts as DRAFT).

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Body field	Type	Required	Rules and notes
<code>metaTitle</code>	string	No	SEO meta title (NFR-12). max length 70
<code>metaDescription</code>	string	No	SEO meta description. max length 170
<code>title</code>	string	Yes	min length 3; max length 200
<code>slug</code>	string	No	Defaults to a slug of the title. max length 160
<code>excerpt</code>	string	No	max length 500
<code>body</code>	array of object	Yes	max items 500
<code>type</code>	enum	No	One of: <code>NEWS</code> , <code>REVIEW</code> , <code>ADVICE</code> , <code>GUIDE</code> , <code>ARTICLE</code>

Body field	Type	Required	Rules and notes
categoryId	string (uuid)	No	
coverImageUrl	string (uri)	No	
featured	boolean	No	
tags	array of string	No	max items 20

GET /api/v1/admin/articles/{id}

One article including drafts.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

PATCH /api/v1/admin/articles/{id}

Update an article.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
metaTitle	string	No	SEO meta title (NFR-12). max length 70
metaDescription	string	No	SEO meta description. max length 170
title	string	No	min length 3; max length 200
slug	string	No	Defaults to a slug of the title. max length 160
excerpt	string	No	max length 500
body	array of object	No	max items 500
type	enum	No	One of: NEWS, REVIEW, ADVICE, GUIDE, ARTICLE
categoryId	string (uuid)	No	
coverImageUrl	string (uri)	No	
featured	boolean	No	
tags	array of string	No	max items 20

DELETE /api/v1/admin/articles/{id}

Delete an article.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 204.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

POST /api/v1/admin/articles/{id}/status

Publish (optionally scheduled), unpublish or archive.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
status	enum	Yes	One of: DRAFT , PUBLISHED , ARCHIVED
publishedAt	string	No	Schedule: publish date (defaults to now)

GET /api/v1/admin/faqs

List all FAQs.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

POST /api/v1/admin/faqs

Create an FAQ.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Body field	Type	Required	Rules and notes
question	string	Yes	min length 3; max length 300
answer	string	Yes	min length 1; max length 5000
category	string	No	max length 60. e.g. " selling "
sortOrder	number	No	
isActive	boolean	No	

PATCH /api/v1/admin/faqs/{id}

Update an FAQ.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
question	string	No	min length 3; max length 300
answer	string	No	min length 1; max length 5000
category	string	No	max length 60. e.g. " selling "
sortOrder	number	No	
isActive	boolean	No	

DELETE /api/v1/admin/faqs/{id}

Delete an FAQ.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 204.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

GET /api/v1/admin/media

List videos and podcasts.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
type	query	enum	No	One of: VIDEO , PODCAST

POST /api/v1/admin/media

Add a video or podcast (starts as DRAFT).

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Body field	Type	Required	Rules and notes
type	enum	Yes	One of: VIDEO, PODCAST
title	string	Yes	min length 3; max length 200
description	string	No	max length 2000
url	string (uri)	Yes	Canonical URL (YouTube, Spotify, ...)
embedUrl	string (uri)	No	
thumbnailUrl	string (uri)	No	
durationSec	number	No	min 0

PATCH /api/v1/admin/media/{id}

Update a video or podcast.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
type	enum	No	One of: VIDEO, PODCAST
title	string	No	min length 3; max length 200
description	string	No	max length 2000
url	string (uri)	No	Canonical URL (YouTube, Spotify, ...)
embedUrl	string (uri)	No	
thumbnailUrl	string (uri)	No	
durationSec	number	No	min 0

DELETE /api/v1/admin/media/{id}

Delete a video or podcast.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 204.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

POST /api/v1/admin/media/{id}/status

Publish, unpublish or archive a video or podcast.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
status	enum	Yes	One of: DRAFT, PUBLISHED, ARCHIVED
publishedAt	string	No	Schedule: publish date (defaults to now)

GET /api/v1/admin/pages

List static pages.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

POST /api/v1/admin/pages

Create a static page.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Body field	Type	Required	Rules and notes
metaTitle	string	No	SEO meta title (NFR-12). max length 70
metaDescription	string	No	SEO meta description. max length 170
slug	string	Yes	min length 2; max length 120. e.g. "privacy-policy"
title	string	Yes	min length 2; max length 200
body	array of object	Yes	max items 500

GET /api/v1/admin/pages/{id}

One static page.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

PATCH /api/v1/admin/pages/{id}

Update a static page.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
metaTitle	string	No	SEO meta title (NFR-12). max length 70
metaDescription	string	No	SEO meta description. max length 170
slug	string	No	min length 2; max length 120. e.g. "privacy-policy"
title	string	No	min length 2; max length 200
body	array of object	No	max items 500

DELETE /api/v1/admin/pages/{id}

Delete a static page.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 204.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

POST /api/v1/admin/pages/{id}/status

Publish or unpublish a static page.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
status	enum	Yes	One of: DRAFT , PUBLISHED , ARCHIVED
publishedAt	string	No	Schedule: publish date (defaults to now)

GET /api/v1/admin/promotions

List promotions.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

POST /api/v1/admin/promotions

Create a promotion.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Body field	Type	Required	Rules and notes
title	string	Yes	min length 3; max length 200
description	string	No	max length 2000
bannerUrl	string (uri)	No	
discountType	enum	No	One of: PERCENT , AMOUNT , NONE
discountValue	number	No	min 0
dealerId	string (uuid)	No	Dealer-specific promotion
startsAt	string	Yes	
endsAt	string	No	
isActive	boolean	No	

PATCH /api/v1/admin/promotions/{id}

Update a promotion.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
title	string	No	min length 3; max length 200
description	string	No	max length 2000
bannerUrl	string (uri)	No	
discountType	enum	No	One of: PERCENT , AMOUNT , NONE
discountValue	number	No	min 0
dealerId	string (uuid)	No	Dealer-specific promotion
startsAt	string	No	
endsAt	string	No	
isActive	boolean	No	

DELETE /api/v1/admin/promotions/{id}

Delete a promotion. Its vehicles are detached.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 204.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

PUT /api/v1/admin/promotions/{id}/vehicles

Set the vehicles included in a promotion.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
vehicleIds	array of string	Yes	max items 500

Admin: newsletter · 2 endpoints

GET /api/v1/admin/newsletter/subscribers

List newsletter subscribers.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
status	query	enum	No	One of: SUBSCRIBED, UNSUBSCRIBED
q	query	string	No	

GET /api/v1/admin/newsletter/subscribers.csv

Download subscribers as a CSV file.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
status	query	enum	No	One of: SUBSCRIBED, UNSUBSCRIBED
q	query	string	No	

Admin: users, statistics, audit, system · 10 endpoints

GET /api/v1/admin/audit-logs

Audit trail: who did what, when, previous and new values.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
entityType	query	string	No	

Parameter	In	Type	Required	Notes
entityId	query	string	No	
actorId	query	string	No	
action	query	string	No	Action prefix, e.g. "vehicle." or "offer.accept"
from	query	string	No	
to	query	string	No	

GET /api/v1/admin/stats

Platform statistics.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

GET /api/v1/admin/system/outbox

Background queue health: depth by status and age of the oldest pending event.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

GET /api/v1/admin/system/outbox/events

Background events by status (default: failed/dead events).

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
status	query	enum	No	One of: PENDING, PROCESSING, DONE, DEAD; default "DEAD"

POST /api/v1/admin/system/outbox/events/{id}/retry

Re-queue a dead-lettered event.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 201.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

GET /api/v1/admin/users

List users. Filter by role, status or text.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
page	query	number	No	min 1; default 1
pageSize	query	number	No	min 1; max 100; default 20
role	query	enum	No	One of: CUSTOMER, DEALER, ADMIN, SUPER_ADMIN
status	query	enum	No	One of: ACTIVE, SUSPENDED, DELETED
q	query	string	No	

POST /api/v1/admin/users

Invite an administrator (super admin only).

Access: Super admin only. **Success:** 201.

Body field	Type	Required	Rules and notes
email	string	Yes	

Body field	Type	Required	Rules and notes
firstName	string	Yes	
lastName	string	Yes	
role	enum	Yes	One of: ADMIN, SUPER_ADMIN

GET /api/v1/admin/users/{id}

One user with dealer membership and activity counts.

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

PATCH /api/v1/admin/users/{id}/role

Change a user's platform role. The user is signed out everywhere.

Access: Super admin only. **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
role	enum	Yes	One of: CUSTOMER, ADMIN, SUPER_ADMIN

PATCH /api/v1/admin/users/{id}/status

Suspend or reactivate a user (suspension signs them out everywhere).

Access: Admin (ADMIN or SUPER_ADMIN). **Success:** 200.

Parameter	In	Type	Required	Notes
id	path	string	Yes	

Body field	Type	Required	Rules and notes
status	enum	Yes	One of: ACTIVE, SUSPENDED
reason	string	No	

9. Current limits and open points

These items are known and planned. App developers should design with them in mind.

Topic	Current state	Impact on apps
Push notifications	Not built. In-app list, email and SMS (provider to be chosen) only.	Poll the unread count. A device-token endpoint will be added for FCM/APNs.
Real-time updates	No WebSockets. Data is fetched on request.	Refresh offer lists on screen open and pull-to-refresh.
SMS gateway	Interface ready, logs messages only.	None until a provider is chosen.
Email verification	Not required at sign-up.	None for now.
Payments	Not in scope. Deals are completed offline by the dealer.	No payment screens.
View counts with CDN caching	Views are counted when the request reaches the server, not on CDN hits.	View numbers are approximate.
Capacity target	The '2 million concurrent users per second' target must be turned into measured targets and load-tested.	None for app code.
Not yet tested on real services	Photo upload to S3/MinIO, image processing, SMTP email and Redis were not run in the test environment.	Test photo uploads on staging first.

9.1 Business decisions waiting for confirmation

- Can a published vehicle be sold directly without being reserved first? (Currently yes.)
- Should reserved vehicles accept new enquiries? (Currently yes, as backup buyers.)
- Default bidding rules: offer validity, changes, withdrawals and counter-offers. (Currently configurable per session; default 48 h, all allowed.)
- Affordability rule: instalment up to 30% of gross income. (Confirm with the finance partner.)
- Automatic valuation: market median, mileage and condition adjustments, 15% trade margin. (Confirm the formula.)

Appendix A. Error codes

Not-found errors use the code **<THING>_NOT_FOUND** with status 404, for example **VEHICLE_NOT_FOUND**, **DEALER_NOT_FOUND**, **OFFER_NOT_FOUND**, **SELL_REQUEST_NOT_FOUND**, **LEAD_NOT_FOUND**. The other codes are:

Code	HTTP	Meaning
BELOW_RESERVE	400	The offer is lower than the reserve price of this bidding session.
COMPARE_COUNT	400	Compare between 2 and 4 variants
COUNTER_TOO_LOW	400	A counter-offer must be higher than the current offer
DEPOSIT_TOO_HIGH	400	Deposit cannot exceed the vehicle price
EMPTY_SEARCH	400	Choose at least one filter to save a search
FILE_TOO_LARGE	400	The file is too big (vehicle and sell photos 15 MB, dealer documents 20 MB).
GENERATION_MODEL_MISMATCH	400	Generation does not belong to this model
HEAD_OFFICE_REQUIRED	400	The head-office branch cannot be deactivated
INVALID_BRANCH	400	Branch does not belong to your dealership
INVALID_CATEGORY	400	One or more categories do not exist
INVALID_CONTENT_BLOCK	400	A content block has an unknown type or invalid fields (details say which block).
INVALID_CURRENT_PASSWORD	400	Current password is incorrect
INVALID_DEALER	400	Dealer not found
INVALID_IDEMPOTENCY_KEY	400	Idempotency-Key is too long
INVALID_MAKE	400	Make not found
INVALID_MEMBER	400	Staff member not found in your dealership
INVALID_PASSWORD	400	Password is incorrect
INVALID_RANGE	400	Expected $estimateLow \leq estimateMid \leq estimateHigh$
INVALID_RESERVATION	400	reservedUntil must be in the future
INVALID_RESET_TOKEN	400	This reset link is invalid or has expired
INVALID_SPECIAL_PRICE	400	specialPrice must be lower than price
INVALID_VEHICLE	400	Vehicle does not belong to your dealership
INVALID_WINDOW	400	closesAt must be after opensAt and in the future
INVALID_YEAR_RANGE	400	yearTo must be on or after yearFrom
MODEL_MAKE_MISMATCH	400	Model does not belong to the selected make
REASON_REQUIRED	400	This action needs a reason: rejecting a listing or dealer, an admin suspension, or marking a lead as lost.
SPEC_REQUIRED	400	transmission and fuelType are required when no catalogue variant provides them
UNKNOWN_IMAGE	400	One or more images do not belong to this vehicle
UNKNOWN_NOTIFICATION_TYPE	400	The notification preferences contain an unknown notification type.
UPLOAD_NOT_FOUND	400	The file has not been uploaded yet
VALIDATION_FAILED	400	Request validation failed (details lists each problem)
VARIANT_MODEL_MISMATCH	400	Variant does not belong to the selected model
WINDOW_TOO_LONG	400	The bidding window is longer than allowed (default maximum 168 hours).
INVALID_CREDENTIALS	401	Email or password is incorrect

Code	HTTP	Meaning
INVALID_REFRESH_TOKEN	401	Refresh token is invalid
INVALID_TOKEN	401	Access token is invalid or expired
REFRESH_TOKEN_EXPIRED	401	Session expired, please sign in again
UNAUTHORIZED	401	Authentication required
ACCOUNT_DISABLED	403	This account is not active
ADMIN_SUSPENSION	403	This listing was suspended by an administrator and can only be reinstated by one
BRANCH_SCOPE	403	You can only add vehicles to your own branch
CANNOT_GRANT	403	You tried to give a staff member a permission you do not have yourself.
DEALER_INACTIVE	403	The dealership is suspended or rejected, so this action is blocked.
DEALER_NOT_APPROVED	403	Your dealership must be approved before you can do this
FORBIDDEN	403	You do not have permission to perform this action
INSUFFICIENT_ROLE	403	Only the owner can manage managers
INVALID_STORAGE_KEY	403	The upload key does not belong to this dealership or request.
LEAD_NOT_ASSIGNED	403	You can only manage leads assigned to you
MISSING_DEALER_PERMISSION	403	Your dealership role does not include the permission this action needs.
NOT_A_DEALER	403	This action requires a dealership account
OWNER_IMMUTABLE	403	The dealership owner cannot be changed here
SELF_MANAGEMENT	403	Admins cannot change their own status or role.
SELF_UPDATE	403	You cannot change your own role or access
SUPER_ADMIN_REQUIRED	403	Only a super admin can manage administrators
NOT_FOUND	404	Record not found
ALREADY_AWARDED	409	An offer has already been accepted for this vehicle
ALREADY_ROUTED	409	This enquiry is already assigned to a dealer
BIDDING_CANCELLED	409	This bidding process was cancelled
BIDDING_CLOSED	409	Bidding is not open for this vehicle
BIDDING_FINISHED	409	This bidding process is finished
BIDDING_IN_PROGRESS	409	This vehicle already has an active bidding process
BRANCH_EXISTS	409	A branch with this name already exists
CANNOT_CANCEL	409	The bidding session is already awarded or cancelled.
CANNOT_REOPEN	409	Only closed sessions without an accepted offer can be reopened.
CONCURRENT_UPDATE	409	Someone changed the record at the same moment. Reload and try again.
COUNTER_NOT_ALLOWED	409	Counter-offers are not enabled for this bidding process
COUNTER_PENDING	409	Wait for the dealer to answer your current counter-offer
DEALER_MEMBER	409	Dealer staff roles are managed by their dealership
DEALER_OWNER	409	Dealership owners must contact support to close the dealership first
EMAIL_TAKEN	409	An account with this email already exists
ENQUIRY_CANCELLED	409	The customer cancelled this enquiry
ENQUIRY_CLOSED	409	This enquiry is closed
FOREIGN_KEY_CONSTRAINT	409	Related record does not exist or is still referenced
IDEMPOTENCY_RETRY	409	Please retry the request

Code	HTTP	Meaning
INVALID_STATE	409	The sell request is in a status that does not allow this action.
INVALID_STATUS_TRANSITION	409	This status change is not allowed from the current status (details.allowed lists the allowed ones). Used for vehicles and dealers.
LIMIT_REACHED	409	The user already has the maximum number of saved searches (25).
MATERIAL_CHANGE_REQUIRES_REVIEW	409	These fields can only change in DRAFT. Move the vehicle back to draft (it will need a new review).
MODIFICATION_NOT_ALLOWED	409	This bidding session does not allow changing bids.
NOT_DEAD	409	Only dead events can be retried
NOT_ELIGIBLE	409	This sell request cannot go to bidding in its current status, or the dealership may not bid on this session.
NO_OPEN_COUNTER	409	There is no open counter-offer to respond to
OFFER_EXPIRED	409	This offer has expired
OFFER_FINAL	409	Your offer is already accepted or superseded and cannot change.
OFFER_NOT_ACTIVE	409	The offer is no longer active.
OFFER_REJECTED	409	This offer was rejected
OFFER_SUPERSEDED	409	This offer was superseded
OFFER_WITHDRAWN	409	The dealer withdrew this offer
REQUEST_IN_PROGRESS	409	A request with this Idempotency-Key is still being processed
SELL_REQUEST_CLOSED	409	This request is closed
STALE_VERSION	409	The record changed since you loaded it (expectedVersion does not match). Reload and try again.
TOO_MANY_IMAGES	409	The maximum number of photos is reached (40 per vehicle, 20 per sell request).
UNIQUE_CONSTRAINT	409	A record with these values already exists
VEHICLE_LOCKED	409	Photos cannot change while the vehicle is pending review, sold or archived.
VEHICLE_NOT_AVAILABLE	409	This vehicle is no longer available for enquiries
VEHICLE_NOT_EDITABLE	409	Vehicles that are pending review, sold or archived cannot be edited.
WITHDRAWAL_NOT_ALLOWED	409	Bids cannot be withdrawn in this bidding process
IDEMPOTENCY_KEY_REUSED	422	Idempotency-Key was already used with a different request
IMAGES_REQUIRED	422	Upload at least one photo before submitting for review
RATE_LIMITED	429	Too many requests, please slow down
INTERNAL_ERROR	500	An unexpected error occurred
INVALID_STAGE_TRANSITION	409	This lead stage change is not allowed.
NOT_FOUND / <THING>_NOT_FOUND	404	The record does not exist, or you are not allowed to see it.

Appendix B. More real responses

These responses were captured from the running backend with demo data. Long values are shortened.

POST /auth/refresh -> HTTP 200

```
{
  "tokenType": "Bearer",
  "accessToken": "<access token>",
  "expiresIn": 900,
  "refreshToken": "<refresh token>",
  "refreshTokenExpiresAt": "2026-11-05T17:46:20.784Z",
  "user": {
    "id": "01a11252-d4d1-743d-bbbb-eca9ba449aaf",
    "email": "docs.1791308780443@example.com",
    "firstName": "Sipho",
    "lastName": "Dlamini",
    "phone": "+27 82 555 0101",
    "role": "CUSTOMER"
  }
}
```

GET /vehicles/does-not-exist (404) -> HTTP 404

```
{
  "statusCode": 404,
  "code": "VEHICLE_NOT_FOUND",
  "message": "Vehicle not found",
  "requestId": "e0ae3d1e-0ddb-4387-be6a-cd99af244014",
  "timestamp": "2026-10-06T17:46:21.451Z",
  "path": "/api/v1/vehicles/does-not-exist"
}
```

POST /dealer/vehicles/{id}/publish on a published vehicle (409) -> HTTP 409

```
{
  "statusCode": 409,
  "code": "INVALID_STATUS_TRANSITION",
  "message": "Cannot move a vehicle from PUBLISHED to PUBLISHED",
  "details": {
    "allowed": [
      "DRAFT",
      "RESERVED"
    ]
  },
  "requestId": "48b7e6e3-9193-4fe2-8444-3e42448bca4d",
  "timestamp": "2026-10-06T17:46:21.467Z",
  "path": "/api/v1/dealer/vehicles/01a11251-678a-706f-a432-6e6ba34717d4/publish"
}
```

Appendix C. Requirement coverage

Where each requirement from the ChangeCars specification is implemented.

Requirement	Topic	Module
FR-01	Registration and authentication	Auth, Customers
FR-02 to FR-07	Search, listings, details, filters and sorting, new and used search	Vehicles, Search, Catalogue
FR-08, FR-09	Sell your vehicle, valuation	Selling
FR-10, FR-52 to FR-56	Dealer bidding, rules, offers, acceptance, expiry, counter-offers	Bidding
FR-11 to FR-13	Dealer registration, vehicle management, dashboard	Dealers, Vehicles, Dealer dashboard
FR-14 to FR-16, FR-20 to FR-22	Contact dealer, quote, Beat My Quote, trade-in, concierge, help me find	Enquiries
FR-17	Vehicle comparison	Vehicles, Catalogue
FR-18, FR-19	Finance and affordability calculators	Finance
FR-23, FR-24	Specials and promotions, hot sellers	Search, Content
FR-25 to FR-27	Categories, make/model/variant catalogue, location search	Catalogue, Search
FR-28 to FR-31, FR-35	Articles, media, insurance and EV information, content management	Content
FR-32	Newsletter	Newsletter
FR-33, FR-42	Enquiry management, customer enquiry tracking	Enquiries
FR-34	Administrator dashboard	Admin and admin endpoints in every module
FR-36, FR-48, FR-57	Notifications and preferences	Notifications plus event handlers
FR-37 to FR-41	Favourites, saved searches, alerts, customer dashboard, recently viewed	Customers, Search
FR-43 to FR-46	Dealer CRM, lead stages, staff, branches	CRM, Dealers
FR-47, FR-49 to FR-51	Vehicle lifecycle, data structure, features, availability	Vehicles, Catalogue
NFR-04, NFR-05, BR-14	Security and role-based access	Auth guards, dealer permission guard
NFR-12	SEO	SEO, slugs, meta fields
NFR-15, BR-13	Logging, monitoring, audit trail	Audit log, structured logs, admin system endpoints
NFR-17	Privacy	Consent fields, data export, account deletion